

6

Alkyl Halides; Nucleophilic Substitution

Goals for Chapter 6

1 Name alkyl halides, explain their physical properties, and describe their common uses.

2 Identify what a substitution is and show how it differs from an elimination.

3 Predict the products of substitution reactions.

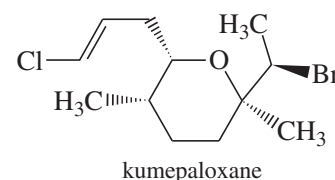
4 Identify the differences between unimolecular and bimolecular substitutions and explain what factors determine the order of the reaction.

5 Given a set of reaction conditions, identify the possible substitution mechanism(s) and predict which product(s) are most likely.

► Halogen-containing organic compounds are not very common in nature. Exceptions occur in ocean plants and invertebrates, where organic halides are common in toxic plants and venoms. The Cymbal bubble snail *Haminoea cymbalum* discharges **kumepaloxane**, a toxic defense chemical containing both chlorine and bromine, when it is disturbed by predatory fishes.



cymbal bubble snail
Haminoea cymbalum



6-1 Introduction

Our study of organic chemistry is organized into families of compounds classified by their functional groups. In this chapter, we consider the properties and reactions of alkyl halides. We use alkyl halides to introduce substitution and elimination, two of the most important types of reactions in organic chemistry. Stereochemistry (Chapter 5) will play a major role in our study of these reactions. Many other reactions show similarities to substitution and elimination, and the techniques introduced in this chapter will be used throughout our study of organic reactions.

There are three major classes of halogenated organic compounds: the alkyl halides, the vinyl halides, and the aryl halides. An **alkyl halide** simply has a halogen atom bonded to one of the sp^3 hybrid carbon atoms of an alkyl group. A **vinyl halide** has a halogen atom bonded to one of the sp^2 hybrid carbon atoms of an alkene. An **aryl halide** has a halogen atom bonded to one of the sp^2 hybrid carbon atoms of an aromatic ring. The chemistry of vinyl halides and aryl halides is different from that of alkyl halides because their bonding and hybridization are different. We consider the reactions of vinyl halides and aryl halides in later chapters. The structures of some representative alkyl halides, vinyl halides, and aryl halides are shown here, with their most common names and uses.

Alkyl halides

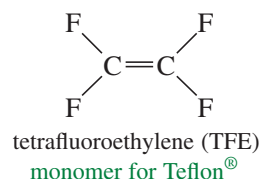
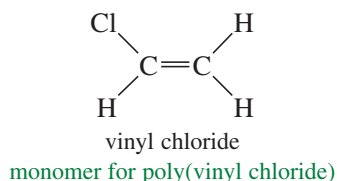
CHCl_3
solvent
chloroform

CHClF_2
refrigerant
Freon-22[®]

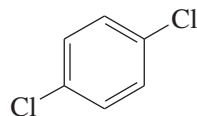
$\text{CCl}_3\text{—CH}_3$
cleaning fluid
1,1,1-trichloroethane

$\text{CF}_3\text{—CHClBr}$
nonflammable anesthetic
Halothane

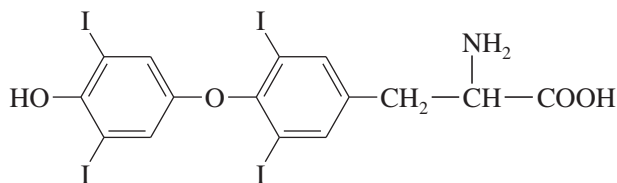
Vinyl halides



Aryl halides

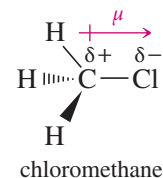


para-dichlorobenzene
mothballs

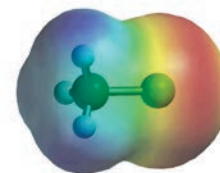


thyroxine
thyroid hormone

The carbon–halogen bond in an alkyl halide is polar because halogen atoms are more electronegative than carbon atoms. Most reactions of alkyl halides result from breaking this polarized bond. The electrostatic potential map of chloromethane (Figure 6-1) shows higher electron density (red) around the chlorine atom and relatively low electron density (blue) around the carbon and hydrogen atoms. The carbon atom has a partial positive charge, making it somewhat electrophilic. A nucleophile can attack this electrophilic carbon, and the halogen atom can leave as a halide ion, taking the bonding pair of electrons with it. By serving as a leaving group, the halogen can be eliminated from the alkyl halide, or it can be replaced (substituted for) by a wide variety of functional groups. This versatility allows alkyl halides to serve as intermediates in the synthesis of many other functional groups.



chloromethane

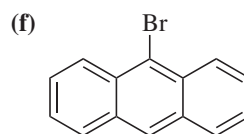
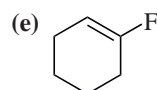
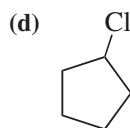


EPM of chloromethane

FIGURE 6-1 Chloromethane and its electrostatic potential map (EPM). The polar C — Cl bond is seen in the EPM as an electron-rich (red) region around chlorine and an electron-poor (blue) region around carbon and the hydrogen atoms.

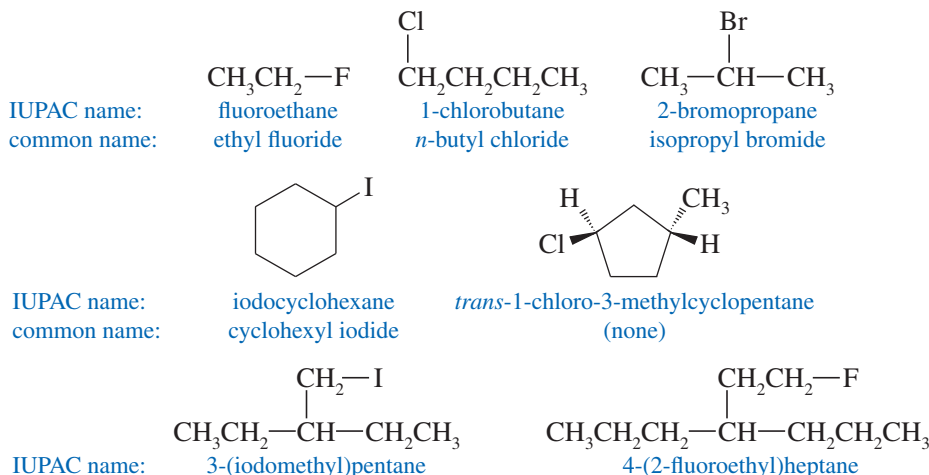
PROBLEM 6-1

Classify each compound as an alkyl halide, a vinyl halide, or an aryl halide.



6-2 Nomenclature of Alkyl Halides

There are two ways of naming alkyl halides. The systematic (IUPAC) nomenclature treats an alkyl halide as an alkane with a *halo-* substituent: Fluorine is *fluoro-*, chlorine is *chloro-*, bromine is *bromo-*, and iodine is *iodo-*. The result is a systematic **haloalkane** name, as in 1-chlorobutane and 2-bromopropane. Common or “trivial” names are constructed by naming the alkyl group and then the halide, as in “isopropyl bromide.” This is the origin of the term *alkyl halide*. Common names are useful only for simple alkyl halides such as the following:



PROBLEM-SOLVING HINT

A summary of organic nomenclature is given in Appendix 5.

Some of the halomethanes have acquired common names that are not clearly related to their structures. A compound of formula CH_2X_2 (a methylene group with two halogens) is called a *methylene halide*, a compound of formula CHX_3 is called a *haloform*, and a compound of formula CX_4 is called a *carbon tetrahalide*.

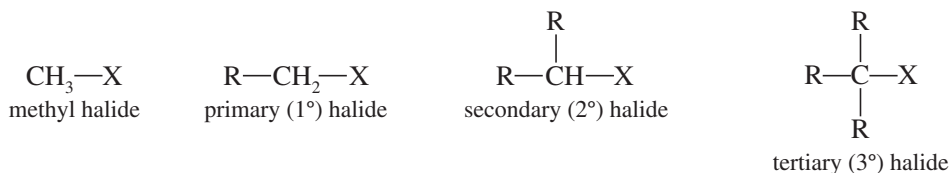
	CH_2Cl_2	CHCl_3	CCl_4
IUPAC name:	dichloromethane	trichloromethane	tetrachloromethane
common name:	methylene chloride	chloroform	carbon tetrachloride

PROBLEM 6-2

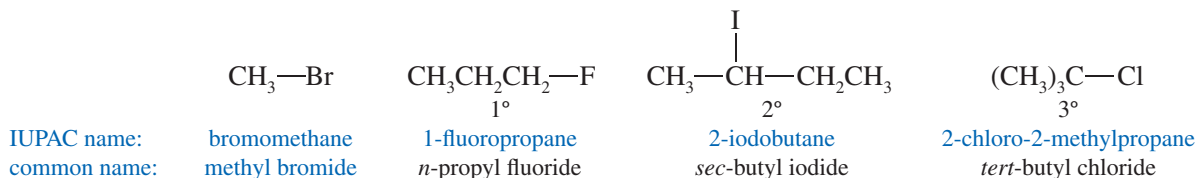
Give the structures of the following compounds.

- | | |
|--|--------------------------------|
| (a) methylene chloride | (b) carbon tetrachloride |
| (c) 3-iodo-2-methylpentane | (d) chloroform |
| (e) 2-chloro-3-ethyl-2-methylhexane | (f) isobutyl iodide |
| (g) <i>cis</i> -1-chloro-3-(chloromethyl)cyclohexane | (h) <i>tert</i> -butyl bromide |

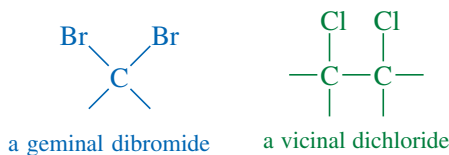
Alkyl halides are classified according to the nature of the carbon atom bonded to the halogen. If the halogen-bearing carbon is bonded to one carbon atom, it is primary (1°) and the alkyl halide is a **primary halide**. If two carbon atoms are bonded to the halogen-bearing carbon, it is secondary (2°) and the compound is a **secondary halide**. A **tertiary halide** (3°) has three other carbon atoms bonded to the halogen-bearing carbon atom. If the halogen-bearing carbon atom is a methyl group (bonded to no other carbon atoms), the compound is a *methyl halide*.



Examples



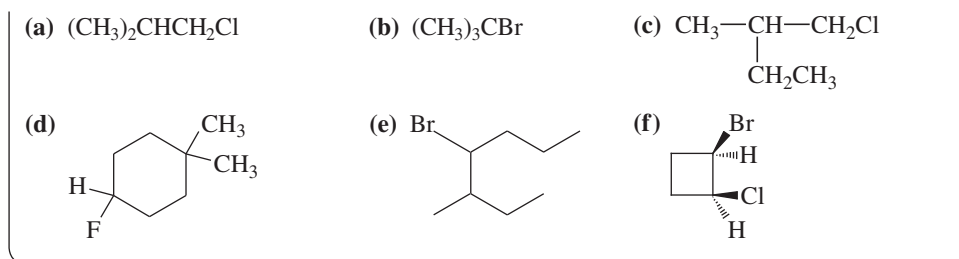
A **geminal dihalide** (Latin, *geminus*, “twin”) has the two halogen atoms bonded to the same carbon atom. A **vicinal dihalide** (Latin, *vicinus*, “neighboring”) has the two halogens bonded to adjacent carbon atoms.



PROBLEM 6-3

For each of the following compounds,

1. give the IUPAC name.
2. give the common name (if possible).
3. classify the compound as a methyl, primary, secondary, or tertiary halide.



6-3 Common Uses of Alkyl Halides

6-3A Solvents

Alkyl halides are used primarily as industrial and household solvents. Carbon tetrachloride (CCl₄) was once used for dry cleaning, spot removing, and other domestic cleaning. Carbon tetrachloride is toxic and carcinogenic (causes cancer), however, so dry cleaners now use 1,1,1-trichloroethane and other solvents instead.

Methylene chloride (CH_2Cl_2) and chloroform (CHCl_3) are also good solvents for cleaning and degreasing work. Methylene chloride was once used to dissolve the caffeine from coffee beans to produce decaffeinated coffee. Concerns about the safety of coffee with residual traces of methylene chloride prompted coffee producers to use supercritical carbon dioxide instead. Chloroform is more toxic and carcinogenic than methylene chloride; it has been replaced by methylene chloride and other solvents in most industrial degreasers and paint removers.

Even the safest halogenated solvents, such as methylene chloride and 1,1,1-trichloroethane, should be used carefully. They are all potentially toxic and carcinogenic, and they dissolve the fatty oils that protect skin, causing a form of dermatitis.

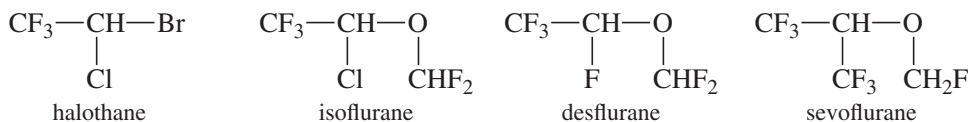
6-3B Reagents

Many syntheses use alkyl halides as starting materials for making more complex molecules. The conversion of alkyl halides to organometallic reagents (compounds containing carbon–metal bonds) is a particularly important tool for organic synthesis. We discuss the formation of organometallic compounds in Section 10-8.

6-3C Anesthetics

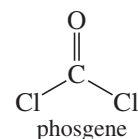
In the 1840s, chloroform (CHCl_3) was found to produce general anesthesia, opening new possibilities for careful surgery with a patient who is unconscious and relaxed. Chloroform is toxic and carcinogenic, however, and it was soon abandoned in favor of safer anesthetics such as diethyl ether. A less toxic halogenated anesthetic is a mixed alkyl halide, CF_3CHClBr , which goes by the trade name Halothane. Ethyl chloride is often used as a topical anesthetic for minor procedures. When sprayed on the skin, its evaporation (bp 12°C) cools the area and enhances the numbing effect.

Chemists have developed other alkyl halides and haloethers that are much less toxic than chloroform, and they are still among the most common anesthetics. Some of the recent ones are Isoflurane, Desflurane, and Sevoflurane. Their structures are compared with Halothane below.



Application: FYI

Chloroform can be converted to phosgene, a very reactive acid chloride, in the presence of oxygen at room temperature. Phosgene is extremely toxic because it reacts with and deactivates many biological molecules. A small amount of alcohol is sometimes added to bottles of chloroform to destroy any phosgene that might be formed.



6-3D Freons: Refrigerants and Foaming Agents

The **freons** (also called *chlorofluorocarbons*, or CFCs) are fluorinated haloalkanes that were developed to replace ammonia as a refrigerant gas. Ammonia is toxic, and leaking refrigerators often killed people who were working or sleeping nearby. Freon-12[®], CF₂Cl₂, was at one time the most widely used refrigerant. Low-boiling freons (such as Freon-11[®], CCl₃F) were once used as *foaming agents* that were added to a plastic to vaporize and form a froth that hardens into a plastic foam.

Freons released into the atmosphere began to destroy Earth's protective ozone layer. CFCs gradually diffuse up into the stratosphere, where the chlorine atoms catalyze the decomposition of ozone (O₃) into oxygen (O₂). Most scientists blame the freon-catalyzed depletion of ozone for the "hole" in the ozone layer that has been detected over the South Pole. Paul Crutzen, Mario Molina, and F. Sherwood Rowland received the 1995 Nobel Prize in Chemistry for their work showing how UV light forms the ozone layer in the atmosphere and how chlorine radicals catalyze its destruction. The opening page of Chapter 4 shows the extent of the ozone hole over the South Pole.

International treaties have limited the future production and use of the ozone-destroying freons. Freon-12 has been replaced in aerosol cans by low-boiling hydrocarbons or carbon dioxide. In refrigerators and automotive air conditioners, Freon-12 has been replaced by Freon-22[®], CHClF₂. Freons with C—H bonds (such as Freon-22), called HCFCs, are generally destroyed at lower altitudes before they reach the stratosphere. Propane, CO₂, and HCFC-123 (CHCl₂CF₃) are used as substitutes for Freon-11 in making plastic foams. Freon-134a (1,1,1,2-tetrafluoroethane), used in automotive air conditioners, contains no chlorine.

Application: Environment

Halogenated compounds persist in the environment because they resist breakdown by soil bacteria. Many are chemically unreactive and insoluble in water, hindering bacterial degradation. Nevertheless, there are strains of bacteria that can use halogenated compounds as their source of food.

6-3E Pesticides

Alkyl halides have contributed to human health through their use as insecticides. Since antiquity, people have died from famine and disease caused or carried by mosquitoes, fleas, lice, and other vermin. The "Black Death" of the Middle Ages wiped out nearly a third of the population of Europe through infection by the flea-borne bubonic plague. Whole regions of Africa and tropical America were uninhabited and unexplored because people could not survive insect-borne diseases such as malaria, yellow fever, and sleeping sickness.

Arsenic compounds, nicotine, and other crude insecticides were developed in the nineteenth century, but these compounds are just as toxic to birds, animals, and people as they are to insects. Their use is extremely hazardous, but a hazardous insecticide was still preferable to certain death by disease or starvation.

The war against insects changed dramatically in 1939 with the discovery of DDT (Figure 6-2). DDT is extremely toxic to insects, but its toxicity in mammals is quite low. About an ounce of DDT is required to kill a person, but that same amount of insecticide protects an acre of land against locusts or mosquitoes. In 1970, the U.S. National Academy of Sciences reported, "in little more than two decades DDT has prevented 500 million deaths due to malaria." Similar advances were made against the mosquitoes carrying yellow fever and the tsetse flies carrying sleeping sickness. Using DDT as a body dust protected people against louse-borne typhus, and dusting rodent burrows controlled the threat of plague.

As with many inventions, DDT showed undesired side effects. It is a long-lasting insecticide, and its residues accumulate in the environment. The widespread use of DDT as an agricultural insecticide led to the development of substantial DDT concentrations in wildlife, causing declines in several species. In 1972, DDT was banned by the U.S. Environmental Protection Agency for use as an agricultural insecticide. It is still used, however, in places where insect-borne diseases threaten human life. DDT-treated bed netting is still the most cost-effective protection against malaria, and careful spraying of DDT around dwellings and in rodent burrows has helped to control the spread of deadly diseases.

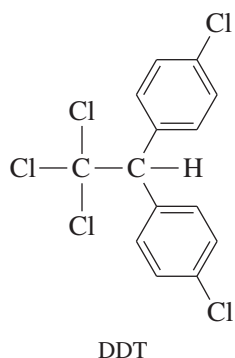


FIGURE 6-2 Structure of DDT. DDT is DichloroDiphenylTrichloroethane, or 1,1,1-trichloro-2,2-bis-(*p*-chlorophenyl) ethane. DDT was the first chlorinated insecticide. Its use rendered large parts of the world safe from insect-borne disease and starvation, but it accumulated in the environment and nearly caused the extinction of several species of birds.

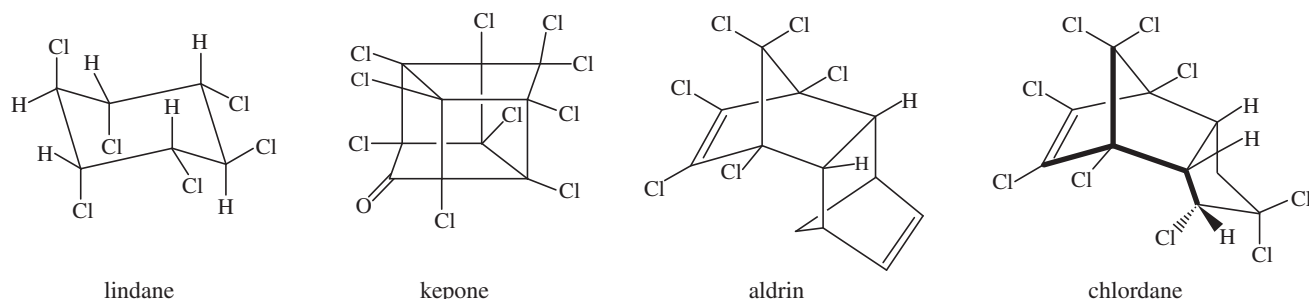
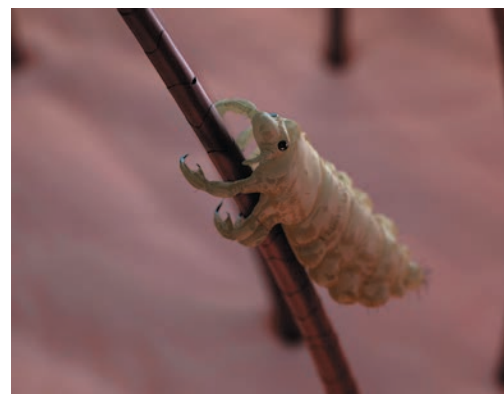
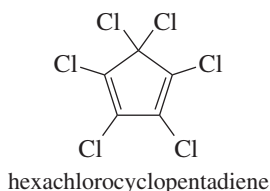


FIGURE 6-3 Four chlorinated insecticides. These pesticides are rarely used because they accumulate in the environment. Lindane is still used in small quantities to control body lice.

Many other chlorinated insecticides have been developed. Most of them also accumulate in the environment, gradually producing toxic effects in wildlife. Others can be used with little adverse impact if they are applied properly. Because of their persistent toxic effects, chlorinated insecticides are rarely used in agriculture. They are generally used when a potent insecticide is needed to protect life or property. For example, lindane is used in shampoos to kill lice, and chlordane was once used to protect wooden buildings from termites. The structures of some chlorinated insecticides are shown in Figure 6-3.

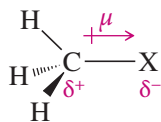
PROBLEM 6-4

Kepone, aldrin, and chlordane are synthesized from hexachlorocyclopentadiene and other five-membered-ring compounds. Show how these three pesticides are composed of two five-membered rings.



6-4 Structure of Alkyl Halides

In an alkyl halide, the halogen atom is bonded to an sp^3 hybrid carbon atom. The halogen is more electronegative than carbon, and the C—X bond is polarized with a partial positive charge on carbon and a partial negative charge on the halogen.



The dipole moment (μ) is given in debyes (D):
 $\mu = 4.8 \times \delta \times d$
 where δ is the amount of charge separation
 and d is the bond length.

The electronegativities of the halogens *increase* in the order

	I	<	Br	<	Cl	<	F
electronegativity:	2.7		3.0		3.2		4.0

The carbon–halogen bond lengths *increase* as the halogen atoms become bigger (larger atomic radii) in the order

	C—F	<	C—Cl	<	C—Br	<	C—I
bond length:	1.38 Å		1.78 Å		1.94 Å		2.14 Å

Body lice transmit a variety of human diseases, including typhus, trench fever, and bubonic plague. Typhus killed millions of people in Europe and Russia during World War II. The epidemic was contained when DDT became readily available for body dusting. Lotions containing lindane (Kwell® and others) are still used as topical treatments for lice.

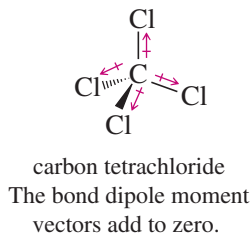
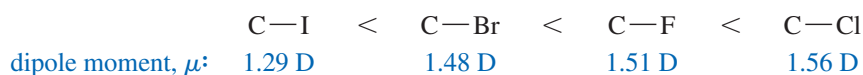


TABLE 6-1 Molecular Dipole Moments of Methyl Halides

X	CH ₃ X	CH ₂ X ₂	CHX ₃	CX ₄
F	1.82 D	1.97 D	1.65 D	0
Cl	1.94 D	1.60 D	1.03 D	0
Br	1.79 D	1.45 D	1.02 D	0
I	1.64 D	1.11 D	1.00 D	0

These two effects oppose each other, with the larger halogens having longer bonds but weaker electronegativities. The overall result is that the bond dipole moments increase in the order



A *molecular* dipole moment is the vector sum of the individual bond dipole moments. Molecular dipole moments are not easy to predict because they depend on the bond angles and other factors that vary with the specific molecule. Table 6-1 lists the experimentally measured dipole moments of the halogenated methanes. Notice how the four symmetrically oriented polar bonds of the carbon tetrahalides cancel to give a molecular dipole moment of zero.

PROBLEM 6-5

For each pair of compounds, predict which one has the higher molecular dipole moment, and explain your reasoning.

- ethyl chloride or ethyl iodide
- 1-bromopropane or cyclopropane
- cis*-2,3-dibromobut-2-ene or *trans*-2,3-dibromobut-2-ene
- cis*-1,2-dichlorocyclobutane or *trans*-1,3-dichlorocyclobutane

6-5 Physical Properties of Alkyl Halides

6-5A Boiling Points

Two types of intermolecular forces influence the boiling points of alkyl halides. The London force (Section 2-2B) is the strongest intermolecular attraction in alkyl halides. London forces are *surface* attractions, resulting from coordinated temporary dipoles. Molecules with larger surface areas have larger London attractions, resulting in higher boiling points. Dipole–dipole attractions (arising from the polar C—X bond) also affect the boiling points, but to a smaller extent.

Molecules with higher molecular weights generally have higher boiling points because they are heavier (and therefore slower moving), and they have greater surface area. The surface areas of the alkyl halides vary with the surface areas of halogens. We can get an idea of the relative surface areas of halogen atoms by considering their van der Waals radii. Figure 6-4 shows that an alkyl fluoride has nearly the same surface area as the corresponding alkane; thus, its London attractive forces are similar. The alkyl fluoride has a larger dipole moment, however, so the total attractive forces are slightly greater in the alkyl fluoride, giving it a higher boiling point. For example, the boiling point of *n*-butane is 0 °C, while that of *n*-butyl fluoride is 33 °C.

The other halogens are considerably larger than fluorine, giving them more surface area and raising the boiling points of their alkyl halides. With a boiling point of 78 °C, *n*-butyl chloride shows the influence of chlorine's much larger surface area. This trend continues with *n*-butyl bromide (bp 102 °C) and *n*-butyl iodide (bp 131 °C). Table 6-2

Halogen	van der Waals Radius (10 ⁻⁸ cm)
F	1.35
Cl	1.8
Br	1.95
I	2.15
H (for comparison)	1.2

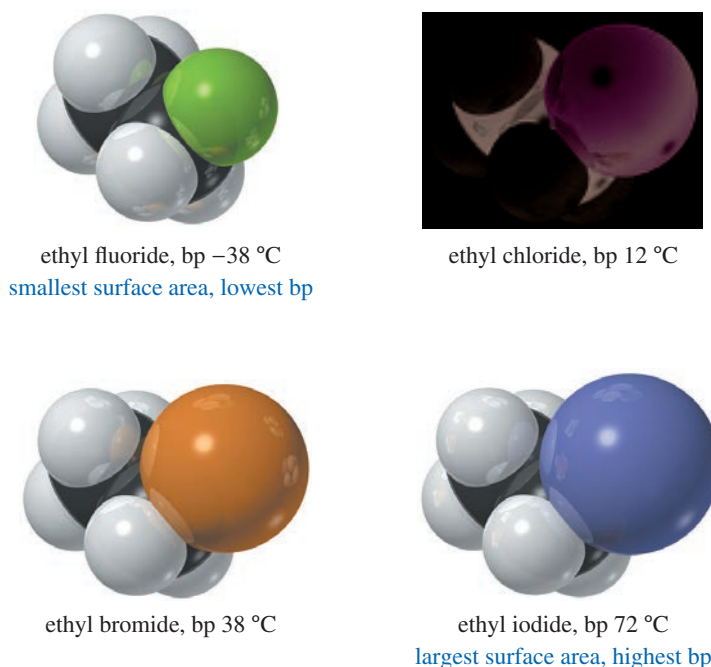


FIGURE 6-4 Space-filling drawings of the ethyl halides. The heavier halogens are larger, with much greater surface areas. As a result, the boiling points of the ethyl halides increase in the order $\text{F} < \text{Cl} < \text{Br} < \text{I}$.

lists the boiling points and densities of some simple alkyl halides. Notice that compounds with branched, more spherical shapes have lower boiling points as a result of their smaller surface areas. For example, *n*-butyl bromide has a boiling point of $102\text{ }^{\circ}\text{C}$, while the more spherical *tert*-butyl bromide has a boiling point of only $73\text{ }^{\circ}\text{C}$. This effect is similar to the one we saw with alkanes.

PROBLEM 6-6

For each pair of compounds, predict which compound has the higher boiling point. Check Table 6-2 to see if your prediction was right; then explain why that compound has the higher boiling point.

- (a) isopropyl bromide and *n*-butyl bromide
- (b) isopropyl chloride and *tert*-butyl bromide
- (c) 1-bromobutane and 1-chlorobutane

6-5B Densities

Table 6-2 also lists the densities of common alkyl halides. Like their boiling points, their densities follow a predictable trend. Alkyl fluorides and alkyl chlorides (those with just one chlorine atom) are less dense than water (1.00 g/mL). Alkyl chlorides with two or more chlorine atoms are denser than water, and all alkyl bromides and alkyl iodides are denser than water.

PROBLEM 6-7

When water is shaken with hexane, the two liquids separate into two phases. Which compound is present in the top phase, and which is present in the bottom phase? When water is shaken with chloroform, a similar two-phase system results. Again, which compound is present in each phase? Explain the difference in the two experiments. What do you expect to happen when water is shaken with ethanol ($\text{CH}_3\text{CH}_2\text{OH}$)?

TABLE 6-2 Physical Properties of Alkyl Halides

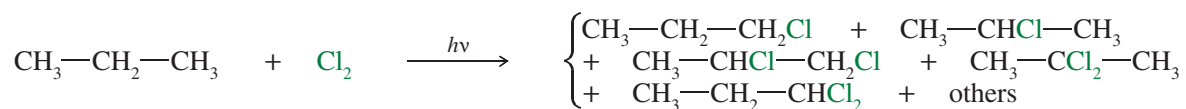
Compound	Molecular Weight	Boiling Point (°C)	Density (g/mL)
CH ₃ —F	34	−78	
CH ₃ —Cl	50.5	−24	0.92
CH ₃ —Br	95	4	1.68
CH ₃ —I	142	42	2.28
CH ₂ Cl ₂	85	40	1.34
CHCl ₃	119	61	1.50
CCl ₄	154	77	1.60
CH ₃ CH ₂ —F	48	−38	0.72
CH ₃ CH ₂ —Cl	64.5	12	0.90
CH ₃ CH ₂ —Br	109	38	1.46
CH ₃ CH ₂ —I	156	72	1.94
CH ₃ CH ₂ CH ₂ —F	62	3	0.80
CH ₃ CH ₂ CH ₂ —Cl	78.5	47	0.89
CH ₃ CH ₂ CH ₂ —Br	123	71	1.35
CH ₃ CH ₂ CH ₂ —I	170	102	1.75
(CH ₃) ₂ CH—Cl	78.5	36	0.86
(CH ₃) ₂ CH—Br	123	59	1.31
(CH ₃) ₂ CH—I	170	89	1.70
CH ₃ CH ₂ CH ₂ CH ₂ —F	76	33	0.78
CH ₃ CH ₂ CH ₂ CH ₂ —Cl	92.5	78	0.89
CH ₃ CH ₂ CH ₂ CH ₂ —Br	137	102	1.28
CH ₃ CH ₂ CH ₂ CH ₂ —I	184	131	1.62
(CH ₃) ₃ C—Cl	92.5	52	0.84
(CH ₃) ₃ C—Br	137	73	1.23
(CH ₃) ₃ C—I	184	100	1.54

6-6 Preparation of Alkyl Halides

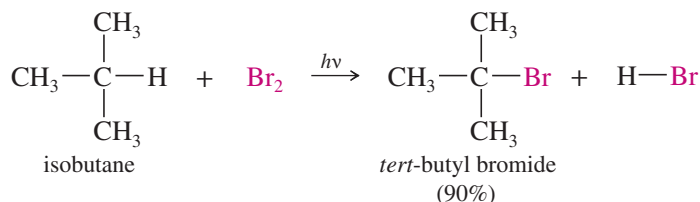
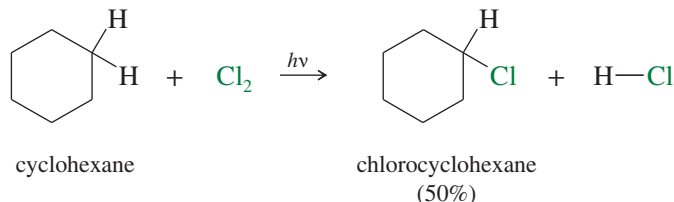
Most syntheses of alkyl halides exploit the chemistry of functional groups we have not yet covered. For now, we review free-radical halogenation and only summarize other, often more useful, syntheses of alkyl halides. The other syntheses are discussed in subsequent chapters.

6-6A Free-Radical Halogenation

Although we discussed its mechanism at length in Section 4-3, free-radical halogenation is rarely an effective method for the synthesis of alkyl halides. It usually produces mixtures of products because there are different kinds of hydrogen atoms that can be abstracted. Also, more than one halogen atom may react, giving multiple substitutions. For example, the chlorination of propane can give a messy mixture of products.



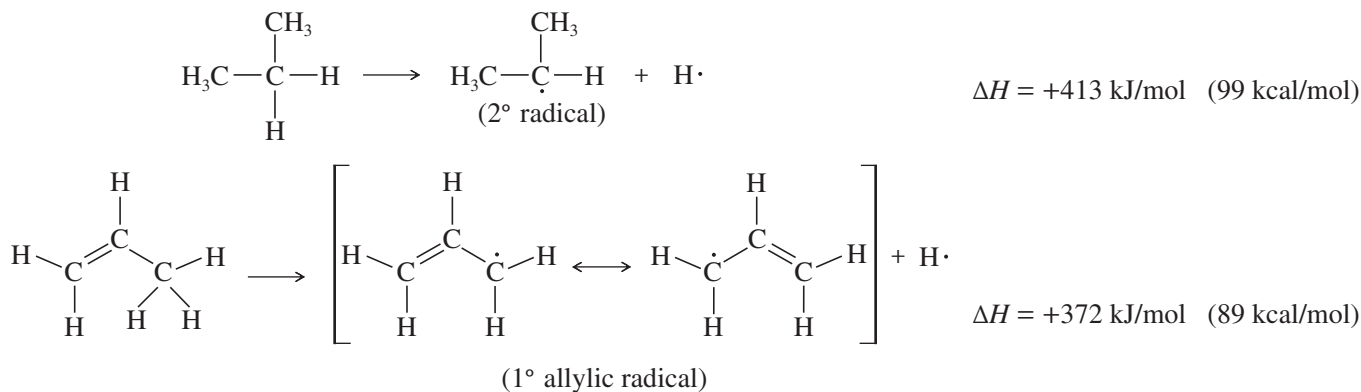
In industry, free-radical halogenation is sometimes useful because the reagents are cheap, the mixture of products can be separated by distillation, and each of the individual products is sold separately. In a laboratory, however, we need a good yield of one particular product. Free-radical halogenation rarely provides good selectivity and yield, so it is seldom used in the laboratory. Laboratory syntheses using free-radical halogenation are generally limited to specialized compounds that give a single major product, such as the following examples.



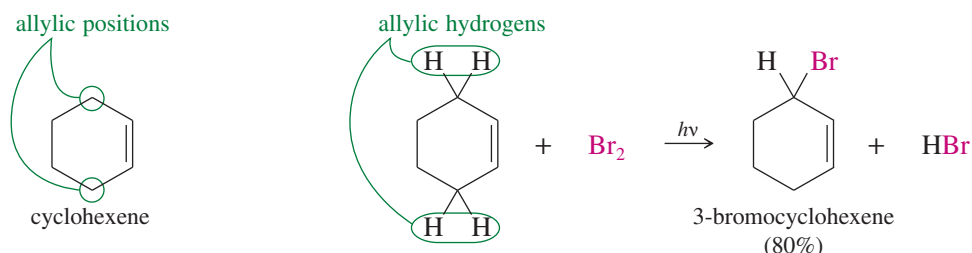
All the hydrogen atoms in cyclohexane are equivalent, and free-radical chlorination gives a usable yield of chlorocyclohexane. Formation of dichlorides and trichlorides is possible, but these side reactions are controlled by using only a small amount of chlorine and an excess of cyclohexane. Free-radical bromination is highly selective (Section 4-14), and it gives good yields of products that have one type of hydrogen atom that is more reactive than the others. Isobutane has only one tertiary hydrogen atom, and this atom is preferentially abstracted to give a tertiary free radical. In general, however, we are not inclined to use free-radical halogenation in the laboratory because it tends to be plagued by mixtures of products.

6-6B Allylic Bromination

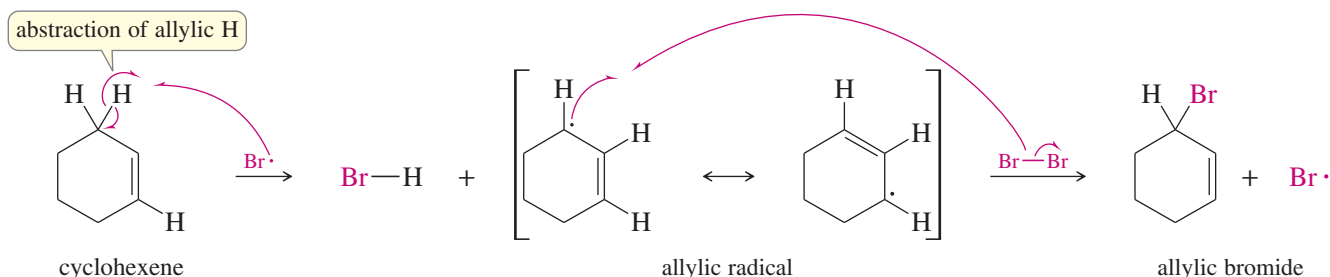
Although free-radical halogenation is a poor synthetic method in most cases, free-radical bromination of alkenes can be carried out in a highly selective manner. An **allylic** position is a carbon atom next to a carbon-carbon double bond. Allylic intermediates (cations, radicals, and anions) are stabilized by resonance with the double bond, allowing the charge or radical to be delocalized. The following bond dissociation enthalpies show that less energy is required to form a resonance-stabilized primary allylic radical than a typical secondary radical.



Recall from Section 4-13C that bromination is highly selective, with only the most stable radical being formed. If there is an allylic hydrogen, the allylic radical is usually the most stable of the radicals that might be formed. For example, consider the free-radical bromination of cyclohexene. Under the right conditions, free-radical bromination of cyclohexene can give a good yield of 3-bromocyclohexene, where bromine has substituted for an allylic hydrogen on the carbon atom next to the double bond.

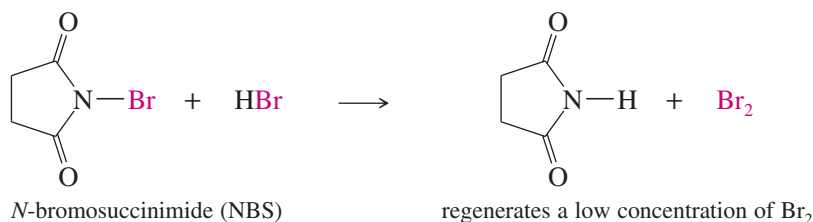


The mechanism is similar to other free-radical halogenations. A bromine radical abstracts an allylic hydrogen atom to give a resonance-stabilized allylic radical. This radical reacts with Br_2 , regenerating a bromine radical that continues the chain reaction.



The general mechanism for allylic bromination shows that either end of the resonance-stabilized allylic radical can react with bromine to give products. In one of the products, the bromine atom appears in the same position where the hydrogen atom was abstracted. The other product results from reaction at the carbon atom that bears the radical in the second resonance form of the allylic radical. This second compound is said to be the product of an **allylic shift**.

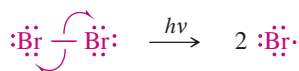
For efficient allylic bromination, a large concentration of bromine must be avoided because bromine can also add to the double bond (Chapter 8). **N-Bromosuccinimide (NBS)** is often used as the bromine source in free-radical brominations because it combines with the HBr by-product to regenerate a constant low concentration of bromine. No additional bromine is needed because most samples of NBS contain traces of Br_2 to initiate the reaction.



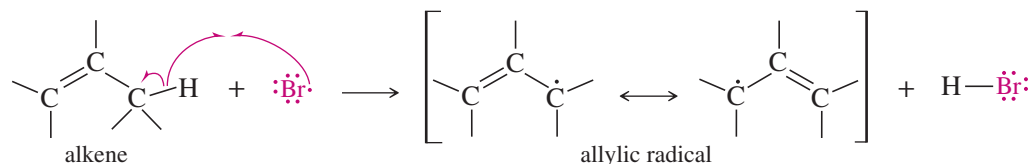
NBS also works well for brominating benzylic positions, next to an aromatic ring (see Problem 6-10). Allylic and benzylic halogenations are discussed in more detail in Chapter 15.

MECHANISM 6-1 Allylic Bromination

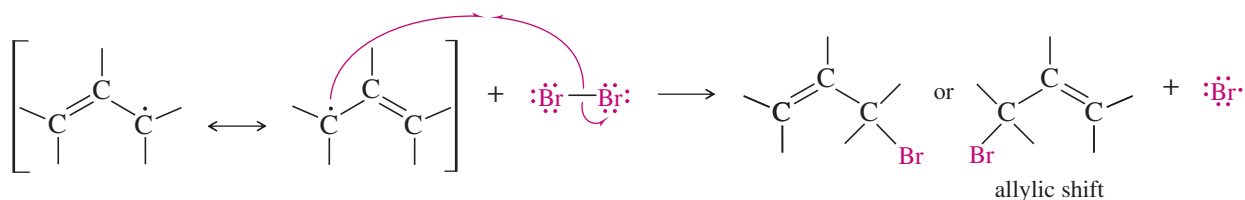
Initiation Step: Bromine absorbs light, causing formation of radicals.



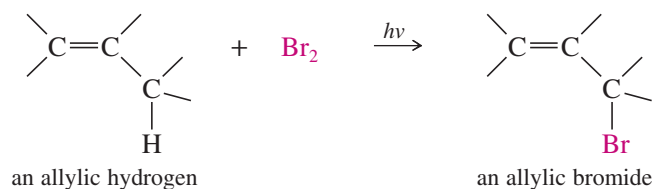
First Propagation Step: A bromine radical abstracts an allylic hydrogen.



Second Propagation Step: Either radical carbon can react with bromine.

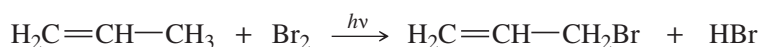


Overall reaction



PROBLEM 6-8

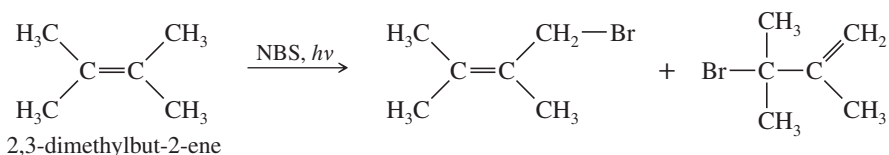
(a) Propose a mechanism for the following reaction:



(b) Use the bond-dissociation enthalpies given in Table 4-2 (page 209) to calculate the value of ΔH° for each step shown in your mechanism. (The BDE for $\text{CH}_2=\text{CHCH}_2-\text{Br}$ is about 280 kJ/mol, or 67 kcal/mol.) Calculate the overall value of ΔH° for the reaction. Are these values consistent with a rapid free-radical chain reaction?

PROBLEM 6-9

The light-initiated reaction of 2,3-dimethylbut-2-ene with *N*-bromosuccinimide (NBS) gives two products:



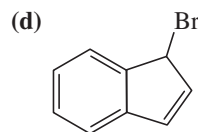
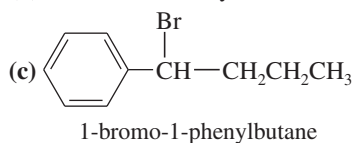
- (a) Give a mechanism for this reaction, showing how the two products arise as a consequence of the resonance-stabilized intermediate.
- (b) The bromination of cyclohexene using NBS gives only one major product, as shown on the previous page. Explain why there is no second product from an allylic shift.

PROBLEM 6-10

Show how free-radical halogenation might be used to synthesize the following compounds. In each case, explain why we expect to get a single major product.

(a) 1-chloro-2,2-dimethylpropane (neopentyl chloride)

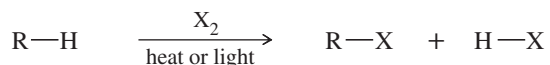
(b) 2-bromo-2-methylbutane



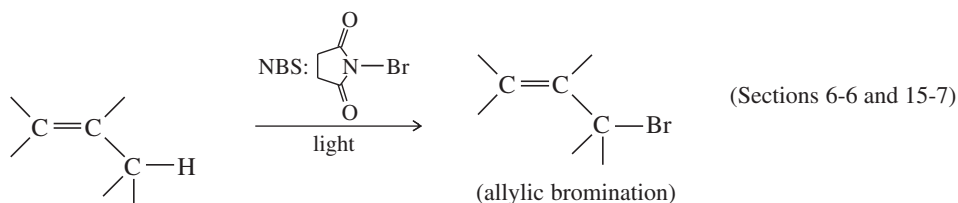
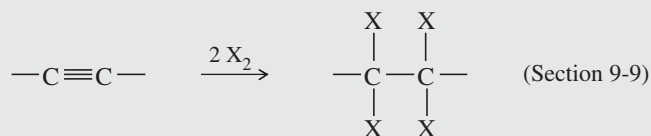
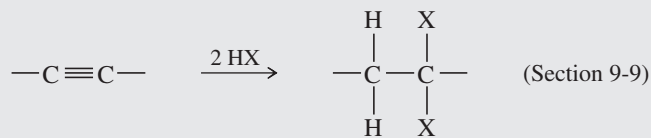
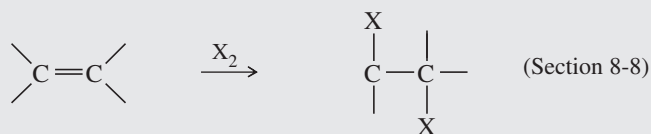
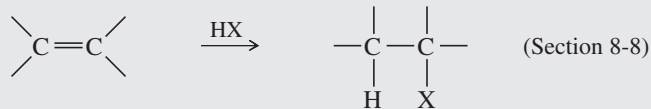
Following is a brief summary of the most important methods of making alkyl halides. Many of them are more general and more useful than free-radical halogenation. Several of these methods are not discussed until later in the text (note the appropriate section references). They are listed here so that you can use this summary for reference throughout the course.

SUMMARY Methods for Preparing Alkyl Halides

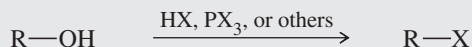
1. From alkanes: free-radical halogenation (synthetically useful only in certain cases) (Sections 4-13 and 6-6)



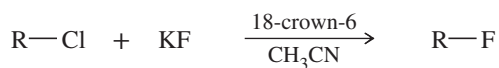
2. From alkenes and alkynes (covered later as shown)



3. From alcohols (covered later in sections 11-7, 11-8 and 11-9)



4. From other halides (Section 6-9)

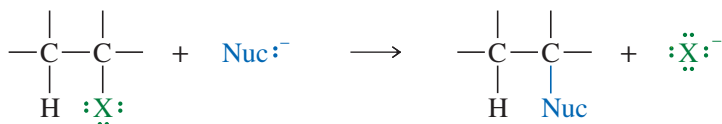


6-7 Reactions of Alkyl Halides: Substitution and Elimination

Alkyl halides are easily converted to many other functional groups. The halogen atom can leave with its bonding pair of electrons to form a stable halide ion; we say that a halide is a good **leaving group**. When another atom replaces the halide ion, the reaction is a **substitution**.

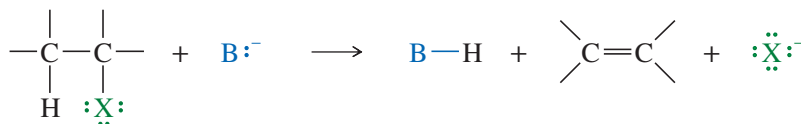
In a **nucleophilic substitution**, a nucleophile (Nuc^-) replaces a leaving group ($:\ddot{\text{X}}^-$) from a carbon atom, using its lone pair of electrons to form a new bond to the carbon atom.

Nucleophilic substitution



When the halide ion leaves along with another atom or ion (often H^+), a new pi bond results, and the reaction is an **elimination**. In many eliminations, a molecule of $\text{H}-\text{X}$ is lost from the alkyl halide to give an alkene. These eliminations are called **dehydrohalogenations** because a hydrogen halide has been removed from the alkyl halide.

Elimination (covered in Chapter 7)

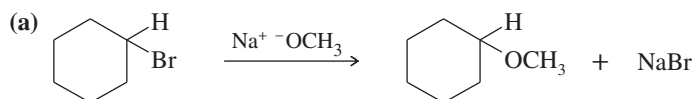


In the elimination (a dehydrohalogenation), the reagent (B^-) reacts as a base, abstracting a proton from the alkyl halide. Most nucleophiles are also basic and can engage in either substitution or elimination, depending on the alkyl halide and the reaction conditions. Therefore, substitution and elimination reactions often compete with each other.

Besides alkyl halides, many other types of compounds undergo substitution and elimination reactions. We introduce substitutions in this chapter and eliminations in Chapter 7, using alkyl halides as our first examples. In later chapters, we encounter substitutions and eliminations of other types of compounds.

PROBLEM 6-11

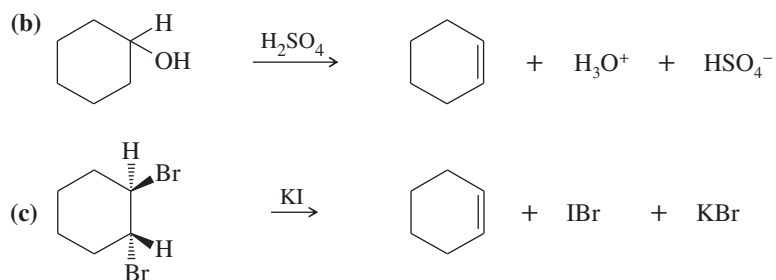
Classify each reaction as a substitution, an elimination, or neither. Identify the leaving group in each reaction, and the nucleophile in substitutions.



(continued)

PROBLEM-SOLVING HINT

Many common nucleophiles are also bases, so the same species might react both as Nuc^- (giving substitution) and B^- (giving elimination) in the same reaction. We study substitutions in Chapter 6, then add eliminations in Chapter 7.

**PROBLEM 6-12**

Give the structures of the substitution products expected when 1-bromohexane reacts with
 (a) $\text{NaOCH}_2\text{CH}_3$ (b) KCN (c) NaOH

6-8 Bimolecular Nucleophilic Substitution: The $\text{S}_{\text{N}}2$ Reaction

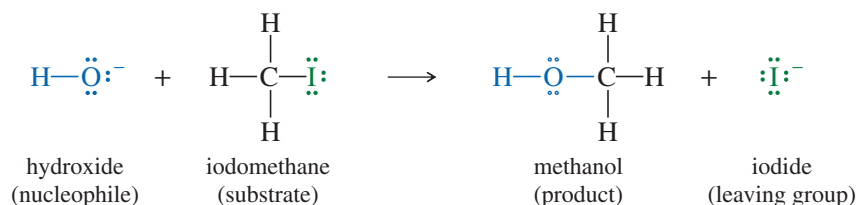
A nucleophilic substitution has the general form

General nucleophilic substitution:

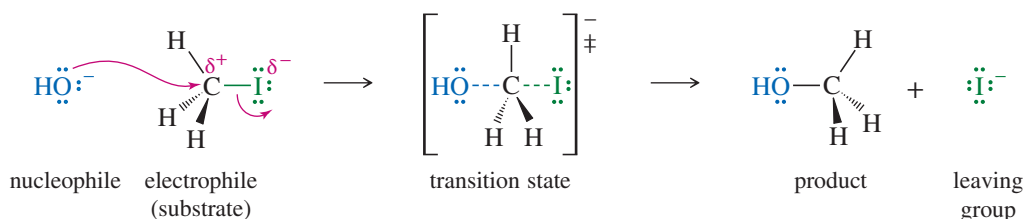


where Nuc^- is the nucleophile and $\ddot{\text{X}}^-$ is the leaving halide ion. An example is the reaction of iodomethane (CH_3I) with hydroxide ion. The product is methanol.

Example:



Hydroxide ion is a strong **nucleophile** (donor of an electron pair) because the oxygen atom has unshared pairs of electrons and a negative charge. Iodomethane is called the **substrate**, meaning the compound that is attacked by the reagent. The carbon atom of iodomethane is electrophilic because it is bonded to an electronegative iodine atom. Electron density is drawn away from carbon by the halogen atom, giving the carbon atom a partial positive charge. The negative charge of hydroxide ion is attracted to this partial positive charge.



Hydroxide ion attacks the back side of the electrophilic carbon atom, donating a pair of electrons to form a new bond. (In general, nucleophiles are said to attack electrophiles, not the other way around.) Notice that curved arrows are used to show the movement of electron pairs—from the electron-rich nucleophile to the electron-poor carbon atom of the electrophile. Carbon can accommodate only eight electrons in its valence shell, so the carbon–iodine bond must begin to break as the carbon–oxygen bond begins to form. Iodide ion is the leaving group; it leaves with the pair of electrons that once bonded it to the carbon atom.

This one-step mechanism is supported by kinetic information. We can vary the concentrations of the reactants and observe the effects on the reaction rate (how much methanol is formed per second). The rate is found to double when the concentration of *either* reactant is doubled. The reaction is therefore first order in each of the reactants and second order overall. The rate equation has the following form:

$$\text{second-order rate} = k_r[\text{CH}_3\text{I}][^-\text{OH}]$$

This rate equation is consistent with a mechanism that requires a collision between a molecule of methyl iodide and a hydroxide ion. Both of these species are present in the transition state, and the collision frequency is proportional to both concentrations. The rate constant k_r depends on several factors, including the energy of the transition state and the temperature (Section 4-9).

This one-step nucleophilic substitution is an example of the **S_N2 mechanism**. The abbreviation S_N2 stands for *Substitution, Nucleophilic, bimolecular*. The term *bimolecular* means that the transition state of the rate-limiting step (the only step in this reaction) involves the collision of *two* molecules. Bimolecular reactions usually have rate equations that are second order overall.

The S_N2 reaction of methyl iodide (iodomethane) with hydroxide ion is a **concerted reaction**, taking place in a single step with bonds breaking and forming at the same time. The middle structure is a **transition state**, a point of maximum energy, rather than an intermediate. In this transition state, the bond to the nucleophile (hydroxide) is partially formed, and the bond to the leaving group (iodide) is partially broken. Remember that a transition state is not a discrete molecule that can be isolated; it exists for only an instant.

The reaction-energy diagram for this substitution (Figure 6-5) shows only one transition state and no intermediates between the reactants and the products. The

PROBLEM-SOLVING HINT

A transition state is unstable and cannot be isolated. It exists for only an instant. In contrast, an intermediate exists for a finite length of time.

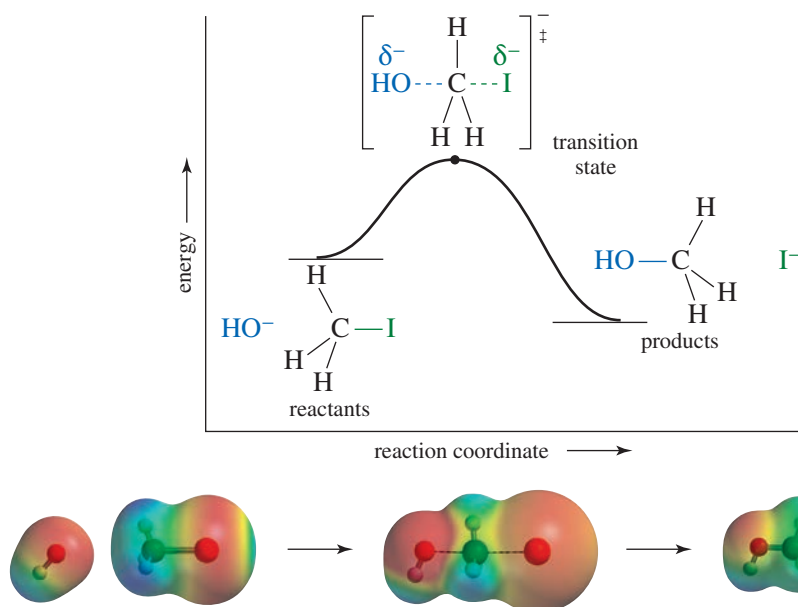


FIGURE 6-5 The reaction-energy diagram for the S_N2 reaction of methyl iodide with hydroxide shows only one energy maximum: the transition state. There are no intermediates.

The electrostatic potential maps of the reactants, transition state, and products show that the negatively charged nucleophile (red) attacks the electrophilic (blue) region of the substrate. In the transition state, the negative charge (red) is delocalized over the nucleophile and the leaving group. The negative charge leaves with the leaving group.

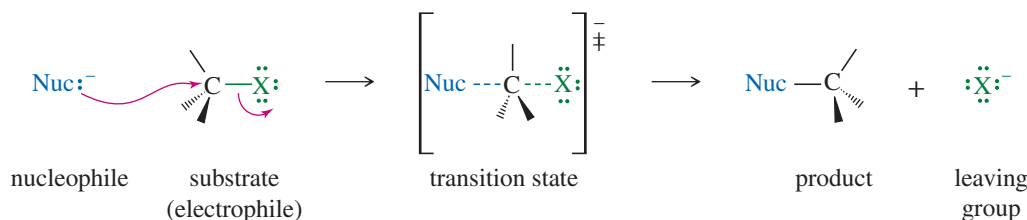
reactants are shown slightly higher in energy than the products because this reaction is known to be exothermic. The transition state is much higher in energy because it involves a five-coordinate carbon atom with two partial bonds.

Key Mechanism 6-2 shows a general S_N2 reaction. A nucleophile attacks the substrate to give a transition state in which a bond to the nucleophile is forming at the same time the bond to the leaving group is breaking.



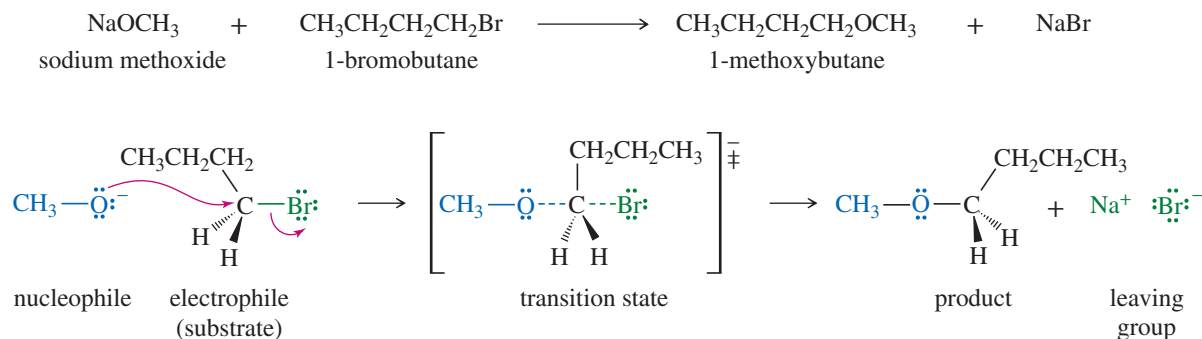
KEY MECHANISM 6-2 The S_N2 Reaction

The S_N2 reaction takes place in a single (concerted) step. A strong nucleophile attacks the electrophilic carbon, forcing the leaving group to leave.



The order of reactivity for substrates is $\text{CH}_3\text{X} > 1^\circ > 2^\circ$. (3° alkyl halides cannot react by this mechanism.)

EXAMPLE: Reaction of 1-bromobutane with sodium methoxide gives 1-methoxybutane.

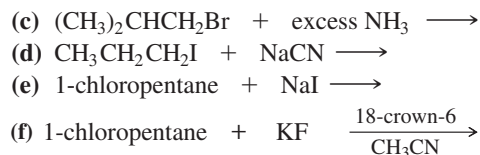


PROBLEM 6-13

- Under certain conditions, the reaction of 0.5 M 1-bromobutane with 1.0 M sodium methoxide forms 1-methoxybutane at a rate of 0.05 mol/L per second. What would be the rate if 0.1 M 1-bromobutane and 2.0 M NaOCH_3 were used?
- Consider the reaction of 1-bromobutane with a large excess of ammonia (NH_3). Draw the reactants, the transition state, and the products. Note that the initial product is the salt of an amine ($\text{RNH}_3^+ \text{Br}^-$), which is deprotonated by the excess ammonia to give the amine.
- Show another S_N2 reaction using a different combination of an alkoxide and an alkyl bromide that also produces 1-methoxybutane.

6-9 Generality of the S_N2 Reaction

Many useful reactions take place by the S_N2 mechanism. The reaction of an alkyl halide, such as methyl iodide, with hydroxide ion gives an alcohol. Other nucleophiles convert alkyl halides to a wide variety of functional groups. The following table summarizes some of the types of compounds that can be formed by nucleophilic displacement of alkyl halides.

**PROBLEM 6-15**

Show how you might use $\text{S}_{\text{N}}2$ reactions to convert 1-chlorobutane into the following compounds.

- (a) butan-1-ol
 (b) 1-fluorobutane
 (c) 1-iodobutane
 (d) $\text{CH}_3-(\text{CH}_2)_3-\text{CN}$
 (e) $\text{CH}_3-(\text{CH}_2)_3-\text{C}\equiv\text{CH}$
 (f) $\text{CH}_3\text{CH}_2-\text{O}-(\text{CH}_2)_3-\text{CH}_3$
 (g) $\text{CH}_3-(\text{CH}_2)_3-\text{NH}_2$

6-10 Factors Affecting $\text{S}_{\text{N}}2$ Reactions: Strength of the Nucleophile

We will use the $\text{S}_{\text{N}}2$ reaction as an example of how we study the factors that affect the rates and products of organic reactions. Both the nucleophile and the substrate (the alkyl halide) are important, as well as the type of solvent used. We begin by considering what makes a good nucleophile.

A “stronger” nucleophile is an ion or a molecule that reacts faster in the $\text{S}_{\text{N}}2$ reaction than a “weaker” nucleophile under the same conditions. A strong nucleophile is much more effective than a weak one in attacking an electrophilic carbon atom. For example, both methanol (CH_3OH) and methoxide ion (CH_3O^-) have easily shared pairs of nonbonding electrons, but methoxide ion reacts with electrophiles in the $\text{S}_{\text{N}}2$ reaction about 1 million times faster than methanol. It is generally true that a species with a negative charge is a stronger nucleophile than a similar, neutral species.

Methoxide ion has nonbonding electrons that are readily available for bonding. In the transition state, the negative charge is shared by the oxygen of methoxide ion and by the halide leaving group. Methanol, however, has no negative charge; the transition state has a partial negative charge on the halide but a partial positive charge on the methanol oxygen atom. We can generalize the case of methanol and the methoxide ion to say that

a base is always a stronger nucleophile than its conjugate acid.

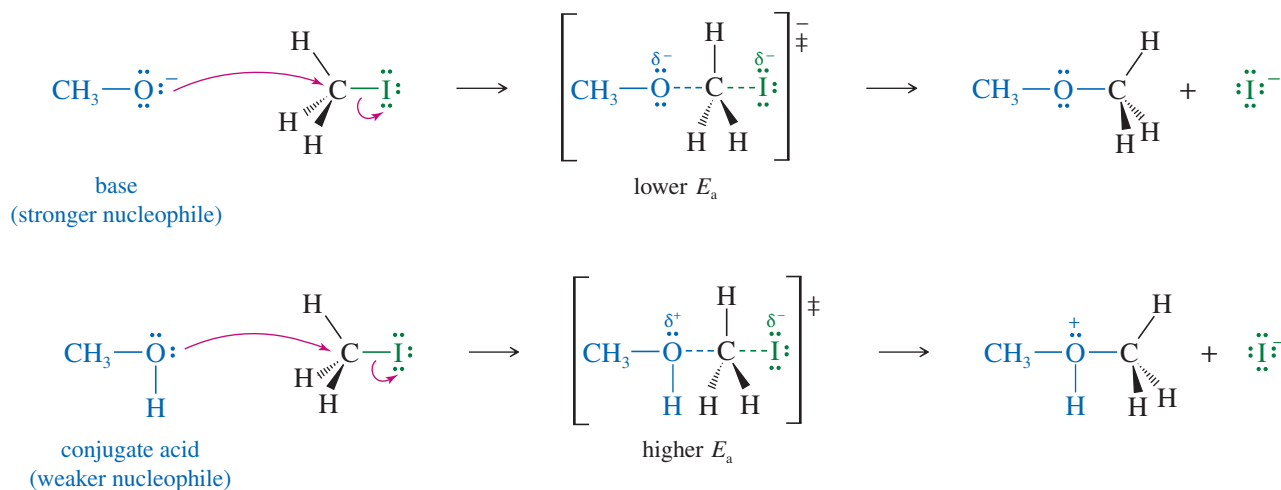


TABLE 6-3 Some Common Nucleophiles

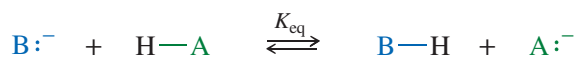
Listed in increasing order of nucleophilicity in polar solvents such as water and alcohols



weak			moderate			strong				
ROH,	H ₂ O	RCOOH	Cl ⁻	RSH,	RNH ₂	RO ⁻ ,	HO ⁻	⁻ CN	I ⁻	NH ₂ ⁻
	F ⁻		RCOO ⁻	RSR	NH ₃	Br ⁻	R ₃ N	RC≡C ⁻	RS ⁻ ,	HS ⁻
									R ₃ P	

We might be tempted to say that methoxide is a much better nucleophile because it is much more basic. This would be a mistake because basicity and nucleophilicity are different properties. **Basicity** is defined by the *equilibrium constant* for abstracting a *proton*. **Nucleophilicity** is defined by the *rate* of attack on an electrophilic *carbon atom*. In both cases, the nucleophile (or base) forms a new bond. If it forms a new bond to a proton, it has reacted as a **base**; if it forms a new bond to carbon, it has reacted as a **nucleophile**. Predicting which way a species will react may be difficult; most (but not all) good nucleophiles are also strong bases, and vice versa.

Basicity: equilibrium for abstracting a proton



Nucleophilicity: rate of attacking a carbon atom

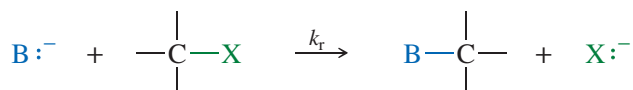


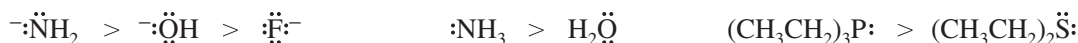
Table 6-3 lists some common nucleophiles in increasing order of their nucleophilicity in polar solvents such as water and alcohols. The strength of nucleophiles in these solvents shows three major trends, as described in the Summary below.

SUMMARY Trends in Nucleophilicity

1. A species with a negative charge is a stronger nucleophile than a similar neutral species. In particular, a base is a stronger nucleophile than its conjugate acid.



2. Nucleophilicity decreases from left to right in the periodic table, following the increase in electronegativity from left to right. The more electronegative elements have more tightly held nonbonding electrons that are less reactive toward forming new bonds.



3. Nucleophilicity increases down the periodic table, following the increase in size and polarizability and the decrease in electronegativity.

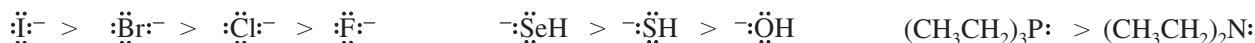
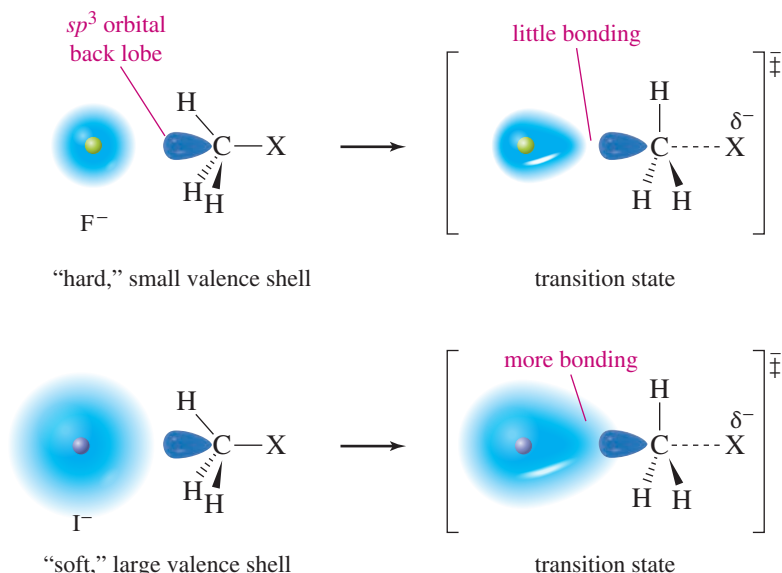
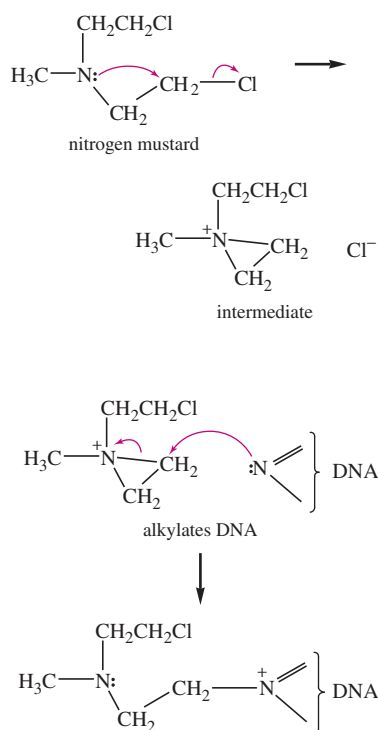


FIGURE 6-6 Comparison of fluoride ion and iodide ion as nucleophiles in the S_N2 reaction. Fluoride has tightly bound electrons that cannot begin to form a C—F bond until the atoms are close together. Iodide has more loosely bound outer electrons that begin bonding earlier in the reaction.



Application: Biochemistry

The “nitrogen mustard” anticancer drugs are believed to alkylate DNA using two S_N2 reactions. First, the nitrogen nucleophile displaces chloride on the primary alkyl chloride portion to generate a reactive intermediate. Second, the intermediate alkylates a nitrogen atom of DNA. The process is repeated, linking the two strands of the double-helix DNA, and thereby preventing replication of the DNA.

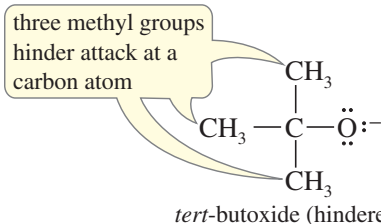


The third trend (size and polarizability) reflects an atom’s ability to engage in partial bonding as it begins to attack an electrophilic carbon atom. As we go down a column in the periodic table, the atoms become larger, with more electrons at a greater distance from the nucleus. The electrons are more loosely held, and the atom is more **polarizable**: Its electrons can move more freely toward a positive charge, resulting in stronger bonding in the transition state. The increased mobility of its electrons enhances the atom’s ability to begin to form a bond at a relatively long distance.

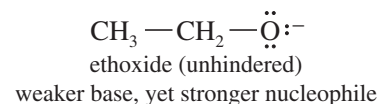
Figure 6-6 illustrates this polarizability effect by comparing the attack of iodide ion and fluoride ion on a methyl halide. The outer shell of the fluoride ion is the second shell. These electrons are tightly held, close to the nucleus. Fluoride is a “hard” (low-polarizability) nucleophile, and its nucleus must approach the carbon nucleus quite closely before the electrons can begin to overlap and form a bond. In the transition state, there is little bonding between fluorine and carbon. In contrast, the outer shell of the iodide ion is the fifth shell. These electrons are loosely held, making the iodide ion a “soft” (high-polarizability) nucleophile. The outer electrons begin to shift and overlap with the carbon atom from farther away. There is a great deal of bonding between iodine and carbon in the transition state, which lowers the energy of the transition state.

6-10A Steric Effects on Nucleophilicity

To serve as a nucleophile, an ion or a molecule must get close to a carbon atom to attack it. Bulky groups on the nucleophile hinder this close approach, and they slow the reaction rate. For example, *tert*-butoxide ion is a stronger *base* (for abstracting protons) than ethoxide ion, but *tert*-butoxide ion has three methyl groups that hinder any close approach to a more crowded carbon atom. Therefore, ethoxide ion is a stronger nucleophile than *tert*-butoxide ion. When bulky groups interfere with a reaction by virtue of their size, we call the effect **steric hindrance**.



tert-butoxide (hindered)
stronger base, yet weaker nucleophile



Steric hindrance has little effect on *basicity* because basicity involves attack on an unhindered proton. When a *nucleophile* attacks a carbon atom, however, a bulky nucleophile cannot approach the carbon atom so easily. Most bases are also nucleophiles, capable of attacking either a proton or an electrophilic carbon atom. If we want a species to act as a base, we use a bulky reagent such as *tert*-butoxide ion. If we want it to react as a nucleophile, we use a less hindered reagent such as ethoxide.

PROBLEM 6-16

For each pair, predict the stronger nucleophile in the S_N2 reaction (using an alcohol as the solvent). Explain your prediction.

- (a) (CH₃CH₂)₃N or (CH₃CH₂)₂NH
 (c) NH₃ or PH₃
 (e) (CH₃)₃N or (CH₃)₂O
 (g) (CH₃)₂CHO[−] or CH₃CH₂CH₂O[−]

- (b) (CH₃)₂O or (CH₃)₂S
 (d) CH₃S[−] or H₂S
 (f) CH₃COO[−] or CF₃COO[−]
 (h) I[−] or Cl[−]

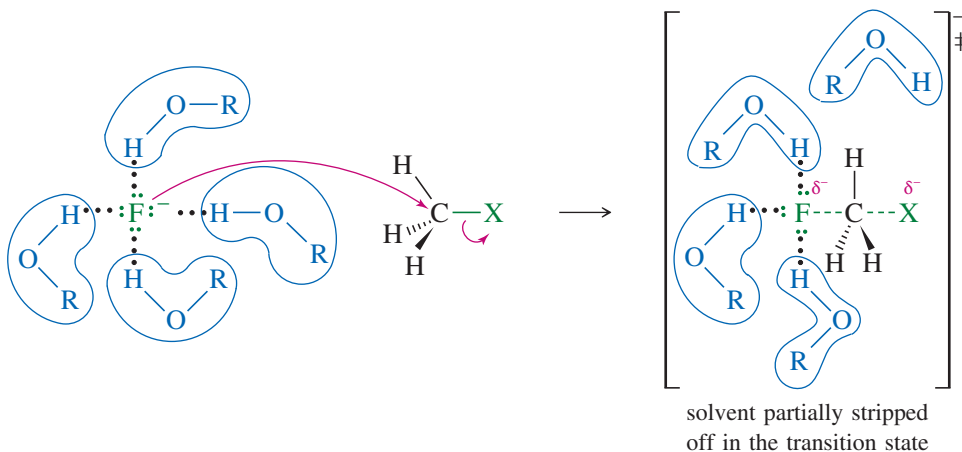
PROBLEM-SOLVING HINT

Steric hindrance (bulkiness) hinders nucleophilicity (S_N2) more than it hinders basicity.

6-10B Solvent Effects on Nucleophilicity

Another factor affecting the nucleophilicity of these ions is their solvation, particularly in protic solvents. A **protic solvent** has acidic protons, usually in the form of O—H or N—H groups. Those with O—H groups are often called **hydroxylic solvents**. These groups form hydrogen bonds to negatively charged nucleophiles. Protic solvents, especially alcohols, are convenient solvents for nucleophilic substitutions because the reagents (alkyl halides, nucleophiles, etc.) tend to be quite soluble.

Small anions are solvated more strongly than large anions in a protic solvent because the solvent molecules approach a small anion more closely and form stronger hydrogen bonds. When an anion reacts as a nucleophile, energy is required to “strip off” some of the solvent molecules, breaking some of the hydrogen bonds that stabilized the solvated anion. More energy is required to strip off solvent from a small, strongly solvated ion such as fluoride than from a large, diffuse, less strongly solvated ion such as iodide.

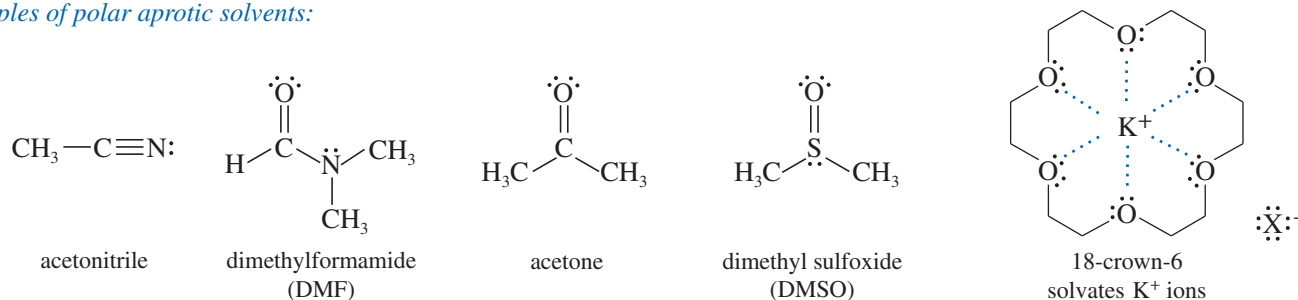


The enhanced solvation of smaller anions in protic solvents, requiring more energy to strip off their solvent molecules, reduces their nucleophilicity. This trend reinforces the trend in polarizability: The polarizability increases with increasing atomic number, and the solvation energy (in protic solvents) decreases with increasing atomic number. Therefore, nucleophilicity (in protic solvents) generally increases down a column in the periodic table, as long as we compare similar species with similar charges.

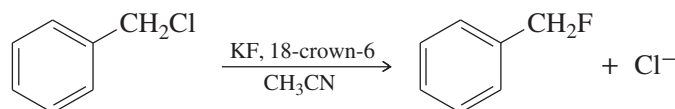
In contrast with protic solvents, **aprotic solvents** (solvents without O—H or N—H groups) enhance the nucleophilicity of anions. An anion is more reactive in an aprotic solvent because it is not so strongly solvated. There are no hydrogen bonds to be broken when solvent must make way for the nucleophile to approach an electrophilic carbon atom. The relatively weak solvating ability of aprotic solvents is also a disadvantage: Most polar, ionic reagents are insoluble in simple aprotic solvents such as alkanes.

Polar aprotic solvents have strong dipole moments to enhance solubility, yet they have no O—H or N—H groups to form hydrogen bonds with anions. Examples of useful polar aprotic solvents (shown below) are acetonitrile, dimethylformamide (DMF), acetone, and dimethyl sulfoxide (DMSO). Changing from a hydroxylic solvent to a polar aprotic solvent often enhances the rate of an S_N2 reaction, sometimes by a factor of 1000 or more. In some cases, we can add specific solvating reagents to enhance solubility without affecting the reactivity of the nucleophile. For example, the “crown ether” 18-crown-6 solvates potassium ions. Using the potassium salt of a nucleophile and solvating the potassium ions causes the nucleophilic anion to be dragged along into solution.

Examples of polar aprotic solvents:



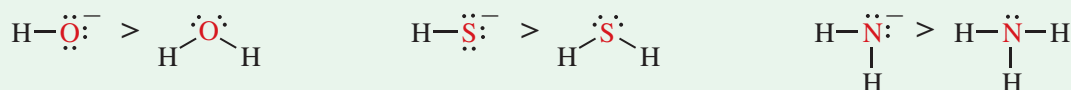
The following example shows how fluoride ion, normally a poor nucleophile in hydroxylic (protic) solvents, can be a good nucleophile in an aprotic solvent. Although KF is not very soluble in acetonitrile, 18-crown-6 solvates the potassium ions, and the poorly solvated fluoride ion follows. This poorly solvated fluoride ion is nucleophilic, and it attacks carbon more readily than it would in a protic solvent.



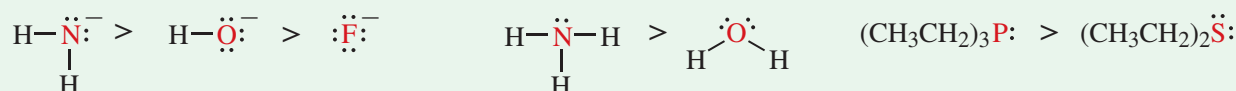
FOCUS Factors Affecting Nucleophilicity

Relative nucleophilicity is a kinetic term, i.e., a better or stronger nucleophile reacts faster at a particular atom.

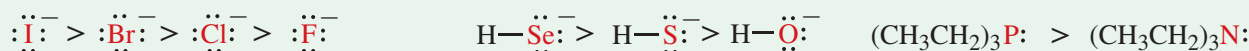
1. A species with a negative charge is a stronger nucleophile than a similar neutral species. In particular, a base is a stronger nucleophile than its conjugate acid.



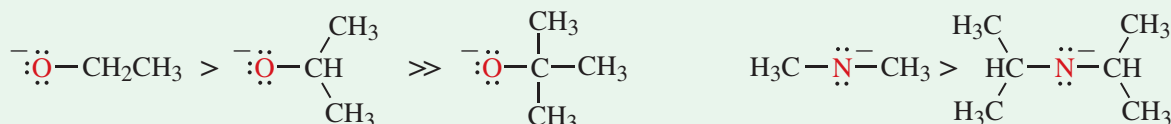
2. Nucleophilicity decreases from left to right across the periodic table, following the increase in electronegativity from left to right. The more electronegative elements have more tightly held nonbonding electrons that are less reactive toward forming new bonds.



3. Nucleophilicity increases down the periodic table, following the increase in size and polarizability and the decrease in electronegativity.

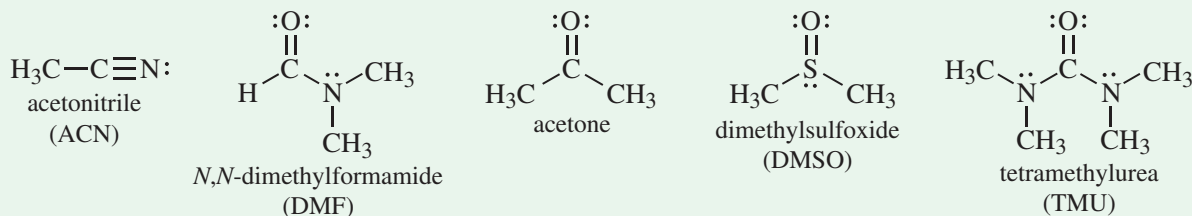


4. Nucleophilicity decreases with increasing steric hindrance around the nucleophilic atom.



5. Nucleophiles are stronger in polar aprotic solvents. Nucleophilicity decreases in protic solvents (any solvent with OH) because hydrogen bonding stabilizes the solvated (surrounded by solvent) nucleophile.

Polar aprotic solvents



6-11 Reactivity of the Substrate in S_N2 Reactions

Just as the nucleophile is important in the S_N2 reaction, the structure of the alkyl halide is equally important. We will often refer to the alkyl halide as the **substrate**: literally, the compound that is being attacked by the reagent. Besides alkyl halides, a variety of other types of compounds serve as substrates in S_N2 reactions. To be a good substrate for S_N2 attack by a nucleophile, a molecule must have an electrophilic carbon atom with a good leaving group, and that carbon atom must not be too sterically hindered for a nucleophile to attack.

6-11A Leaving-Group Effects on the Substrate

A leaving group serves two purposes in the S_N2 reaction:

1. It polarizes the C—X bond, making the carbon atom electrophilic.
2. It leaves with the pair of electrons that once bonded it to the electrophilic carbon atom.

To fill these roles, a good leaving group should be

1. electron-withdrawing, to polarize the carbon atom,
2. stable (not a strong base) once it has left, and
3. polarizable, to stabilize the transition state.

1. The leaving group must be *electron-withdrawing* to create a partial positive charge on the carbon atom, making the carbon electrophilic. An electron-withdrawing leaving group also stabilizes the negatively charged transition state. Halogen atoms are strongly electronegative, so alkyl halides are common substrates for S_N2 reactions. Oxygen, nitrogen, and sulfur also form strongly polarized bonds with carbon; given the right substituents, they can form the basis for excellent leaving groups.

Strongly polarized



2. The leaving group must be *stable* once it has left with the pair of electrons that bonded it to carbon. A stable leaving group is needed for favorable energetics. The leaving group is leaving in the transition state; a reactive leaving group would raise the energy of the transition state, slowing the reaction. Also, the energy of the leaving group is reflected in the energy of the products. A reactive leaving group would raise the energy of the products, driving the equilibrium toward the reactants.

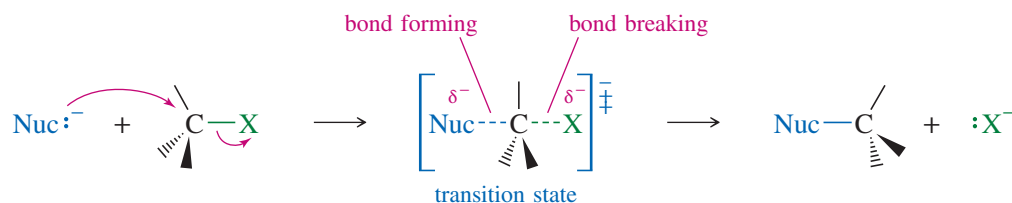


TABLE 6-4 Weak Bases That are Common Leaving Groups

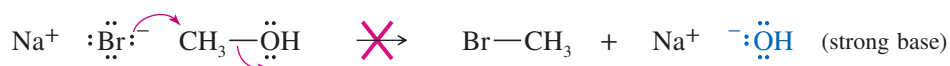
Ions:	$\text{:}\ddot{\text{Cl}}\text{:}^-$	$\text{:}\ddot{\text{Br}}\text{:}^-$	$\text{:}\ddot{\text{I}}\text{:}^-$	$\text{:}\ddot{\text{O}}\text{--}\overset{\overset{\text{:}\ddot{\text{O}}\text{:}}{\parallel}}{\underset{\underset{\text{:}\ddot{\text{O}}\text{:}}{\parallel}}{\text{S}}}\text{--R}$	$\text{:}\ddot{\text{O}}\text{--}\overset{\overset{\text{:}\ddot{\text{O}}\text{:}}{\parallel}}{\underset{\underset{\text{:}\ddot{\text{O}}\text{:}}{\parallel}}{\text{S}}}\text{--}\ddot{\text{O}}\text{R}$	$\text{:}\ddot{\text{O}}\text{--}\overset{\overset{\text{:}\ddot{\text{O}}\text{:}}{\parallel}}{\underset{\underset{\text{:}\ddot{\text{O}}\text{:}}{\parallel}}{\text{P}}}\text{--}\ddot{\text{O}}\text{R}$
	halides			sulfonate	sulfate	phosphate
Neutral molecules:	$\text{H}\bracket{\text{:}\ddot{\text{O}}\text{--H}}$	$\text{H}\bracket{\text{:}\ddot{\text{O}}\text{--R}}$	$\text{R}\bracket{\text{:}\ddot{\text{N}}\text{--R}}\bracket{\text{R}}$	$\text{R}\bracket{\text{:}\ddot{\text{S}}\text{--R}}$		
	water	alcohols	amines	sulfides		

PROBLEM-SOLVING HINT

Strong bases make poor leaving groups. Do not write $\text{S}_{\text{N}}2$ reactions that show hydroxide ions, alkoxide ions, or other strong bases serving as leaving groups.

Good leaving groups should be *weak bases*; therefore, they are the conjugate bases of strong acids. The hydrohalic acids HCl, HBr, and HI are strong, and their conjugate bases (Cl^- , Br^- , and I^-) are all weak bases. Other weak bases, such as sulfate ions, sulfonate ions, and phosphate ions, can also serve as good leaving groups. Table 6-4 lists examples of good leaving groups.

Hydroxide ions, alkoxide ions, and other strong bases are poor leaving groups for $\text{S}_{\text{N}}2$ reactions. For example, the --OH group of an alcohol is a poor leaving group because it would have to leave as hydroxide ion.



Ions that are strong bases and poor leaving groups:

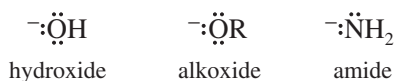
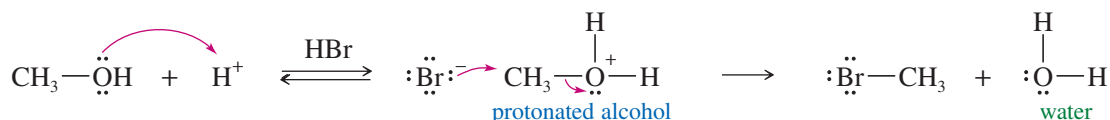


Table 6-4 also lists some neutral molecules that can be good leaving groups. A neutral molecule often serves as the leaving group from a *positively charged* species. For example, if an alcohol is placed in an acidic solution, the hydroxy group is protonated. Water then serves as the leaving group. Note that the need to protonate the alcohol (requiring acid) limits the choice of nucleophiles to those few that are weak bases, such as bromide and iodide. A strongly basic nucleophile would become protonated in acid.

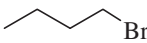
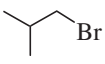


3. Finally, a good leaving group should be *polarizable* to maintain partial bonding with the carbon atom in the transition state. This bonding helps stabilize the transition state and reduce the activation energy. The departure of a leaving group is much like the attack of a nucleophile, except that the bond is breaking rather than forming. Polarizable nucleophiles and polarizable leaving groups both stabilize the transition state by engaging in more bonding at a longer distance. Iodide ion, one of the most polarizable ions, is both a good nucleophile and a good leaving group. In contrast, fluoride ion is a small, “hard” ion. Fluoride is both a poor nucleophile (in protic solvents) and a poor leaving group in $\text{S}_{\text{N}}2$ reactions.

PROBLEM 6-17

When diethyl ether ($\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$) is treated with concentrated HBr, the initial products are $\text{CH}_3\text{CH}_2\text{Br}$ and $\text{CH}_3\text{CH}_2\text{OH}$. Propose a mechanism to account for this reaction.

TABLE 6-5 Effect of Substituents on the Rates of S_N2 Reaction

Class of Halide	Example	Structure	Relative Rate
methyl	CH ₃ —Br		>1000
primary (1°)	CH ₃ CH ₂ —Br		50
secondary (2°)	(CH ₃) ₂ CH—Br		1
tertiary (3°)	(CH ₃) ₃ C—Br		<0.001
<i>n</i> -butyl (1°)	CH ₃ CH ₂ CH ₂ CH ₂ —Br		20
isobutyl (1°)	(CH ₃) ₂ CHCH ₂ —Br		2
neopentyl (1°)	(CH ₃) ₃ CCH ₂ —Br		0.0005

Note: Two or three alkyl groups, or even a single bulky alkyl group, slow the reaction rate. The rates listed are compared to the secondary case (isopropyl bromide), assigned a relative rate of 1.

6-11B Steric Effects on the Substrate

Different alkyl halides undergo S_N2 reactions at vastly different rates. The structure of the substrate is the most important factor in its reactivity toward S_N2 displacement. The reaction goes rapidly with methyl halides and with most primary substrates. It is more sluggish with secondary halides. Tertiary halides fail to react at all by the S_N2 mechanism. Table 6-5 shows the effect of alkyl substitution on the rate of S_N2 displacements.

For simple alkyl halides, the relative rates for S_N2 displacement are



The physical explanation for this order of reactivity is suggested by the information in Table 6-5. All the slow-reacting compounds have one property in common: The back side of the electrophilic carbon atom is crowded by the presence of bulky groups. Tertiary halides are more hindered than secondary halides, which are more hindered than primary halides. Even a bulky primary halide (like neopentyl bromide) undergoes S_N2 reaction at a rate similar to that of a tertiary halide. The relative rates show that it is the bulk of the alkyl groups, rather than an electronic effect, that hinders the reactivity of bulky alkyl halides in the S_N2 displacement.

This effect on the rate is another example of **steric hindrance**. When the nucleophile approaches the back side of the electrophilic carbon atom, it must come within bonding distance of the back lobe of the C—X *sp*³ orbital. If two alkyl groups are bonded to the carbon atom, this process is difficult. Three alkyl groups make it impossible. Just one alkyl group can produce a large amount of steric hindrance if it is unusually bulky, such as the *tert*-butyl group of neopentyl bromide.

Figure 6-7 shows the S_N2 reaction of hydroxide ion with ethyl bromide (1°), isopropyl bromide (2°), and *tert*-butyl bromide (3°). The nucleophile can easily approach the electrophilic carbon atom of ethyl bromide. In isopropyl bromide, the approach is hindered, but still possible. In contrast, S_N2 approach to the tertiary carbon of *tert*-butyl bromide is impossible because of the steric hindrance of the three methyl groups. Make models of ethyl bromide, isopropyl bromide, and *tert*-butyl bromide, and compare the ease of bringing in an atom for a back-side attack.

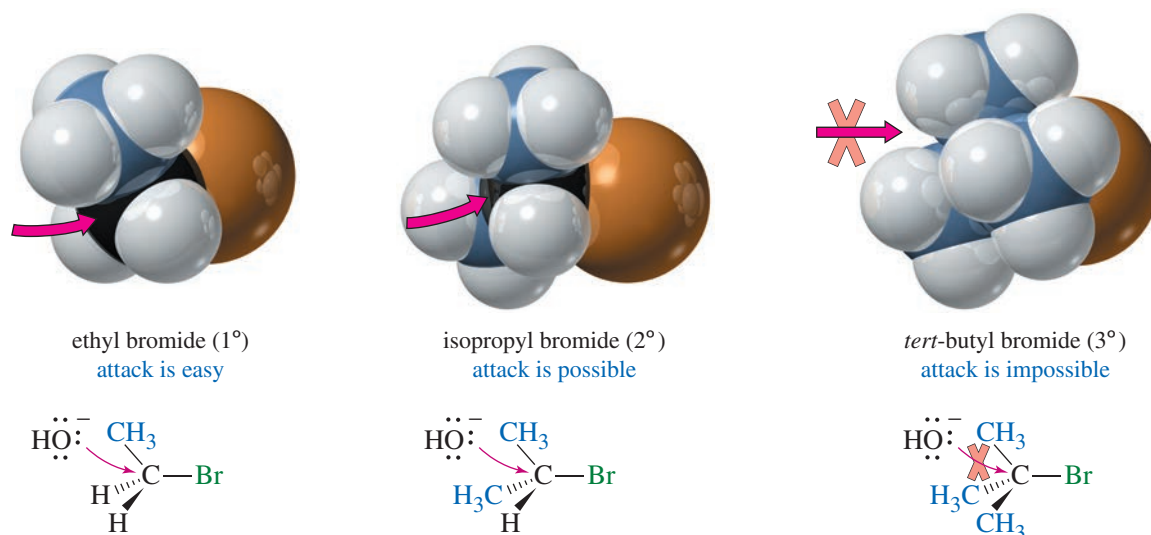


FIGURE 6-7 S_N2 attack on a simple primary alkyl halide is unhindered. Attack on a secondary halide is hindered, and attack on a tertiary halide is impossible.

PROBLEM-SOLVING HINT

Do not write S_N2 reactions occurring on tertiary alkyl halides.

PROBLEM 6-18

Rank the following compounds in decreasing order of their reactivity toward the S_N2 reaction with sodium ethoxide ($\text{Na}^+ \text{ } ^-\text{OCH}_2\text{CH}_3$) in ethanol.

methyl chloride	<i>tert</i> -butyl iodide	neopentyl bromide
isopropyl bromide	methyl iodide	ethyl chloride

PROBLEM 6-19

For each pair of compounds, state which compound is the better S_N2 substrate.

- 2-methyl-1-iodopropane or *tert*-butyl iodide
- cyclohexyl bromide or 1-bromo-1-methylcyclohexane
- 2-bromobutane or isopropyl bromide
- 1-chloro-2,2-dimethylbutane or 2-chlorobutane
- 1-iodobutane or 2-iodopropane

6-12 Stereochemistry of the S_N2 Reaction

As we have seen, the S_N2 reaction requires attack by a nucleophile on the back side of an electrophilic carbon atom (Figure 6-8). A carbon atom can have only four filled bonding orbitals (an octet), so the leaving group must leave as the nucleophile bonds to the carbon atom. The nucleophile's electrons insert into the back lobe of carbon's sp^3 hybrid orbital in its antibonding combination with the orbital of the leaving group (because the bonding MO is already filled). These electrons in the antibonding MO help to weaken the $\text{C}-\text{Br}$ bond as bromide leaves. The transition state shows partial bonding to both the nucleophile and the leaving group.

Back-side attack literally turns the tetrahedron of the carbon atom inside out, like an umbrella caught by the wind (Figure 6-8). In the product, the nucleophile assumes a stereochemical position opposite the position the leaving group originally occupied. We call this result an **inversion of configuration** at the carbon atom.

In the case of an asymmetric carbon atom, back-side attack gives the opposite configuration of the carbon atom (Mechanism 6-3). The S_N2 displacement is the most

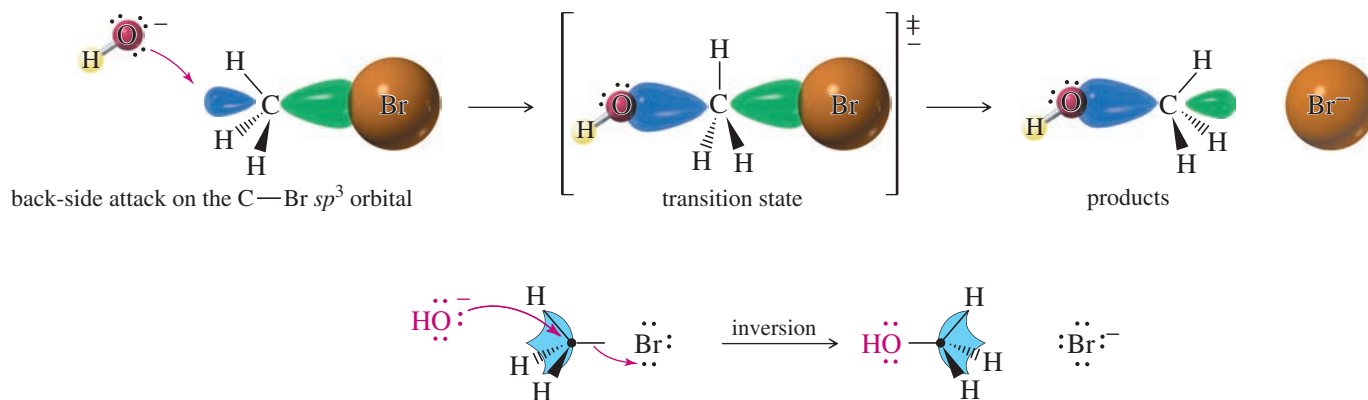
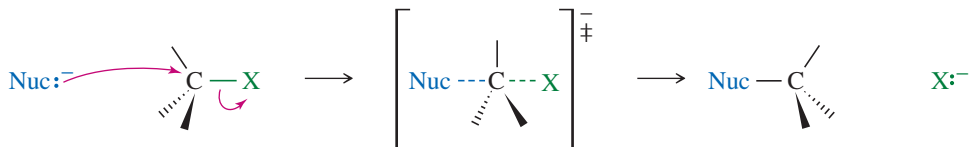


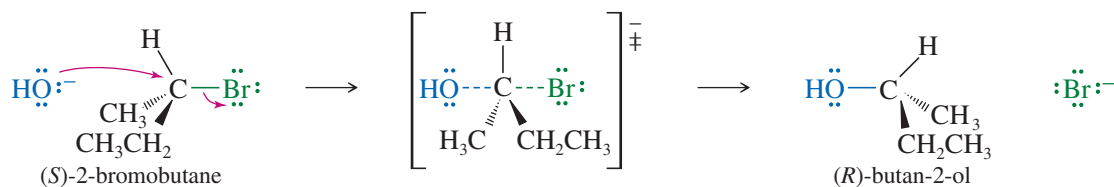
FIGURE 6-8 Back-side attack in the S_N2 reaction. The S_N2 reaction takes place through nucleophilic attack on the back lobe of carbon's sp^3 hybrid orbital. This back-side attack inverts the carbon atom's tetrahedron, like a strong wind inverts an umbrella.

MECHANISM 6-3 Inversion of Configuration in the S_N2 Reaction

Back-side attack inverts the configuration of the carbon atom.

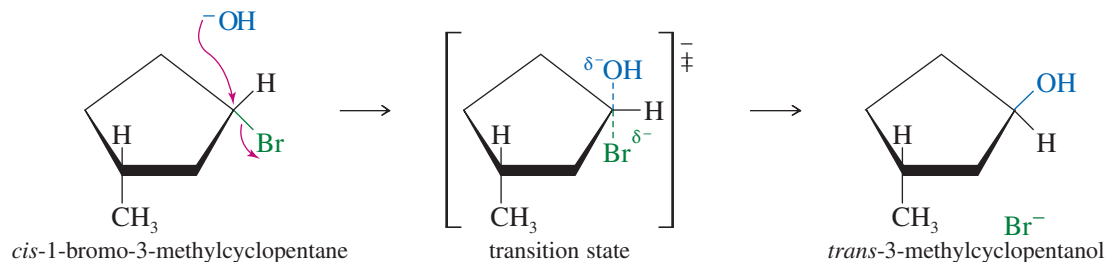


EXAMPLE:



common example of a **Walden inversion**, a step (in a reaction sequence) where an asymmetric carbon atom undergoes inversion of configuration. In the 1890s, Paul Walden, of the University of Tübingen (Germany), was one of the first to study reactions giving inversion of configuration.

In some cases, inversion of configuration is readily apparent. For example, when *cis*-1-bromo-3-methylcyclopentane undergoes S_N2 displacement by hydroxide ion, inversion of configuration gives *trans*-3-methylcyclopentanol.



The S_N2 displacement is a good example of a **stereospecific reaction**: one in which different stereoisomers react to give different stereoisomers of the product. To study

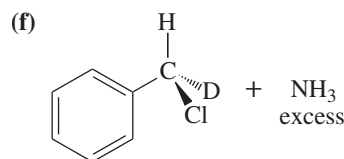
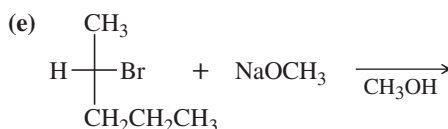
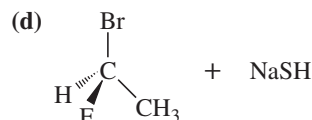
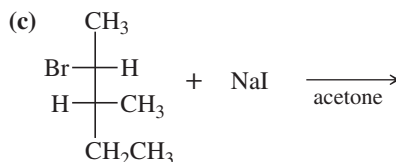
the mechanism of a nucleophilic substitution, we often look at the product to see if the reaction is stereospecific, with inversion of configuration. If it is, the S_N2 mechanism is a good possibility, especially if the reaction kinetics are second order. In many cases (no asymmetric carbon or ring, for example), it is impossible to determine whether inversion has occurred. In these cases, we use kinetics and other evidence to help determine the reaction mechanism.

PROBLEM 6-20

Draw a perspective structure or a Fischer projection for the products of the following S_N2 reactions.

(a) *trans*-1-bromo-3-methylcyclopentane + KOH

(b) (*R*)-2-bromopentane + KCN



PROBLEM-SOLVING HINT

(*R*) and (*S*) are just names, and they vary with changing priorities of the substituents. Don't rely on names to determine the stereochemistry of a reaction.

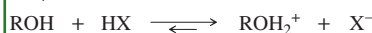
PROBLEM-SOLVING HINT

Strong acids such as HCl, HBr, and HI are completely dissociated in solutions of alcohols or water.

Consider the pK_a values:

HI	−10
HBr	−9
HCl	−7
CH ₃ OH ₂ ⁺	−2.5
CH ₃ CH ₂ OH ₂ ⁺	−2.4
H ₃ O ⁺	−1.7

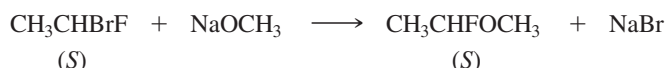
These pK_a values show that the equilibrium



lies far to the right. When we write "HBr (aq)" or "HBr in ethanol," we actually mean the dissociated ions. The values also show that alcohols and water are much more basic than halide ions, so they are more likely than halide ion to serve as the base in the final deprotonation in the S_N1 reaction and similar reactions under acidic conditions.

PROBLEM 6-21

Under appropriate conditions, (*S*)-1-bromo-1-fluoroethane reacts with sodium methoxide to give pure (*S*)-1-fluoro-1-methoxyethane.



(a) Why is bromide rather than fluoride replaced?

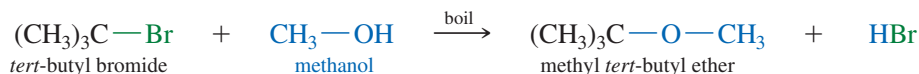
(b) Draw perspective structures (as shown on the previous page for 2-bromobutane) for the starting material, the transition state, and the product.

(c) Does the product show retention or inversion of configuration?

(d) Is this result consistent with reaction by the S_N2 mechanism?

6-13 Unimolecular Nucleophilic Substitution: The S_N1 Reaction

When *tert*-butyl bromide is placed in boiling methanol, methyl *tert*-butyl ether can be isolated from the reaction mixture. Because this reaction takes place with the solvent acting as the nucleophile, it is called a **solvolysis** (*solvo* for "solvent," plus *lysis*, meaning "cleavage").



This solvolysis is a substitution because methoxide has replaced bromide on the *tert*-butyl group. It does not go through the S_N2 mechanism, however. The S_N2 requires a strong nucleophile and a substrate that is not too hindered. Methanol is a weak nucleophile, and *tert*-butyl bromide is a hindered tertiary halide—a poor S_N2 substrate.

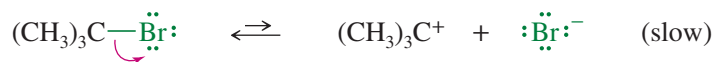
If this substitution cannot go by the S_N2 mechanism, what kind of mechanism might be involved? An important clue is kinetic: Its rate does not depend on the concentration of methanol, the nucleophile. The rate depends only on the concentration of the substrate, *tert*-butyl bromide.

$$\text{first-order rate} = k_r[(\text{CH}_3)_3\text{C}-\text{Br}]$$

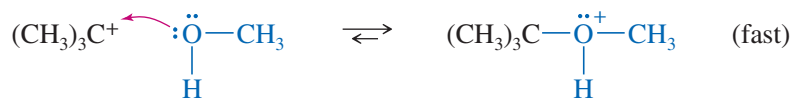
This rate equation is first order overall: first order in the concentration of the alkyl halide and zeroth order in the concentration of the nucleophile. Because the rate does not depend on the concentration of the nucleophile, we infer that the nucleophile is not present in the transition state of the rate-limiting step. The nucleophile must react *after* the slow step.

This type of substitution is called an **S_N1 reaction**, for *Substitution, Nucleophilic, unimolecular*. The term *unimolecular* means there is only one molecule involved in the transition state of the rate-limiting step. The mechanism of the S_N1 reaction of *tert*-butyl bromide with methanol is shown here. Ionization of the alkyl halide (first step) is the rate-limiting step.

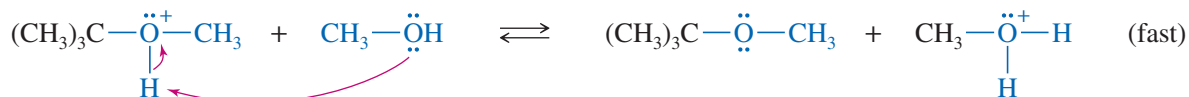
Step 1: Formation of carbocation (rate limiting)



Step 2: Nucleophilic attack on the carbocation



Final Step: Loss of proton to solvent



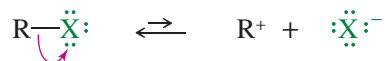
The S_N1 mechanism is a multistep process. The first step is a slow ionization to form a carbocation. The second step is a fast attack on the carbocation by a nucleophile. The carbocation is a strong electrophile; it reacts very quickly with nucleophiles, including weak nucleophiles. The nucleophile in S_N1 reactions is usually weak, because a strong nucleophile would be more likely to attack the substrate and force some kind of second-order reaction. If the nucleophile is an uncharged molecule such as water or an alcohol, the positively charged product must lose a proton to give the final uncharged product. The general mechanism for the S_N1 reaction is summarized in Key Mechanism 6-4.



KEY MECHANISM 6-4 The S_N1 Reaction

The S_N1 reaction involves a two-step mechanism. A slow ionization gives a carbocation that reacts quickly with a (usually weak) nucleophile. Reactivity: 3° > 2° > 1°.

Step 1: Formation of the carbocation (rate-limiting).



Step 2: Nucleophilic attack on the carbocation (fast).

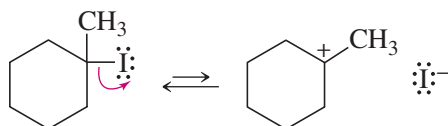


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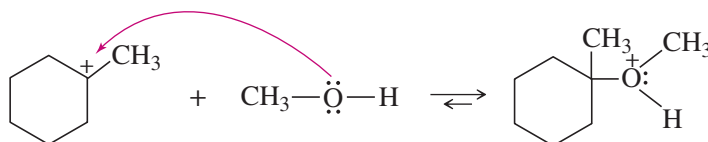
If the nucleophile is water or an alcohol, a third step is needed to deprotonate the product.

EXAMPLE: Solvolysis of 1-iodo-1-methylcyclohexane in methanol.

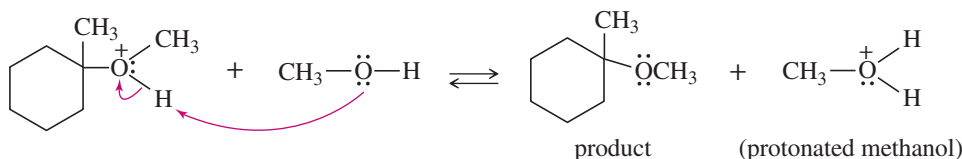
Step 1: Formation of a carbocation (rate-limiting).



Step 2: Nucleophilic attack by the solvent (methanol).



Step 3: Deprotonation to form the product.



PROBLEM 6-22

Propose an S_N1 mechanism for the solvolysis of 3-bromo-2,3-dimethylpentane in ethanol.

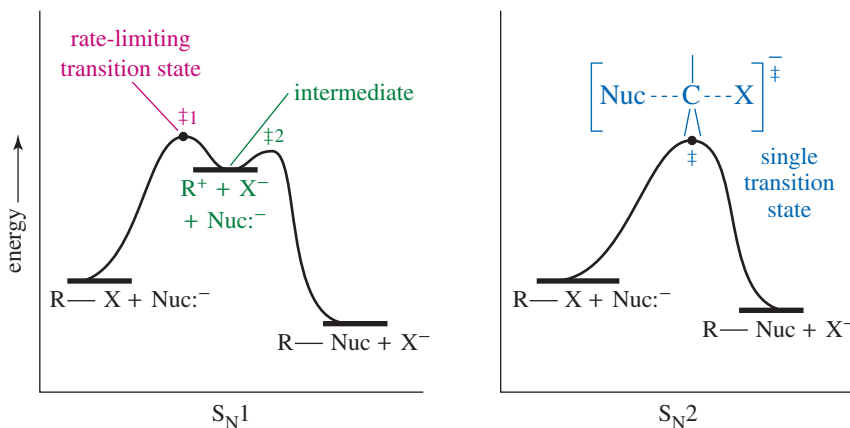
PROBLEM-SOLVING HINT

Never show a proton falling off into thin air. Show a possible base (often the solvent) abstracting the proton.

The reaction-energy diagram of the S_N1 reaction (Figure 6-9) shows why the rate does not depend on the strength or concentration of the nucleophile. The ionization (first step) is highly endothermic, and its large activation energy determines the overall reaction rate. The nucleophilic attack (second step) is strongly exothermic, with a lower-energy transition state. In effect, a nucleophile reacts with the carbocation almost as soon as it forms.

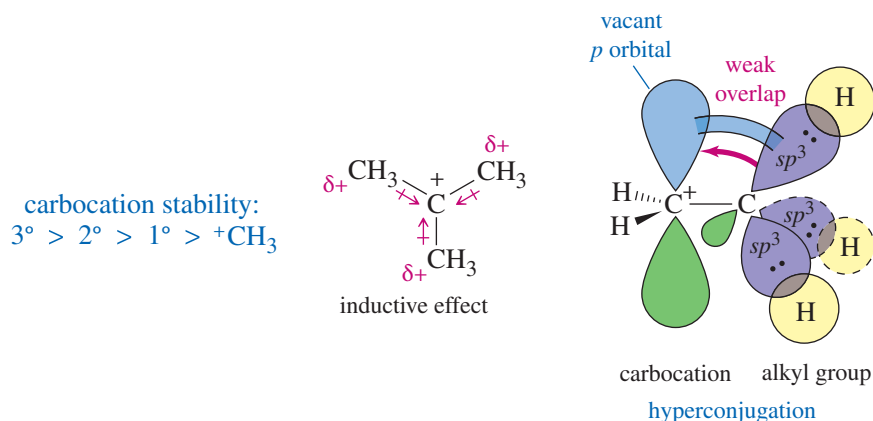
The reaction-energy diagrams of the S_N1 mechanism and the S_N2 mechanism are compared in Figure 6-9. The S_N1 has a true intermediate, the carbocation. The intermediate appears as a relative minimum (a low point) in the reaction-energy diagram. Reagents and conditions that favor formation of the carbocation (the slow step) accelerate the S_N1 reaction; reagents and conditions that hinder its formation retard the reaction.

FIGURE 6-9 Reaction-energy diagrams of the S_N1 and S_N2 reactions. The S_N1 is a two-step mechanism with two transition states ($\ddagger_1$ and $\ddagger_2$) and a carbocation intermediate. The S_N2 has only one transition state and no intermediate.

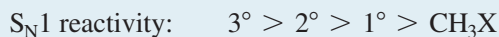


6-13A Substituent Effects

The rate-limiting step of the S_N1 reaction is ionization to form a carbocation, a strongly endothermic process. The first transition state resembles the carbocation (Hammond's postulate, Section 4-14); consequently, rates of S_N1 reactions depend strongly on carbocation stability. In Section 4-16A, we saw that alkyl groups stabilize carbocations by donating electrons through sigma bonds (the *inductive effect*) and through overlap of filled orbitals with the empty *p* orbital of the carbocation (*hyperconjugation*). Highly substituted carbocations are therefore more stable.

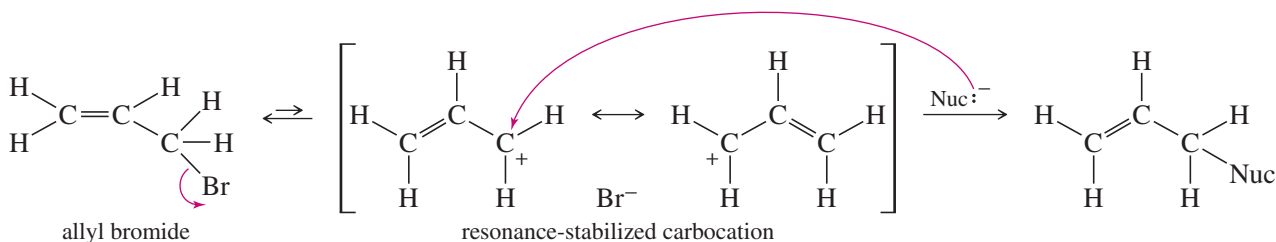


Reactivity toward S_N1 substitution mechanisms follows the stability of carbocations:

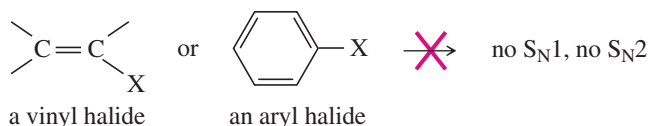


This order is *opposite* that of the S_N2 reaction. Alkyl groups hinder the S_N2 by blocking attack of the strong nucleophile, but alkyl groups enhance the S_N1 by stabilizing the carbocation intermediate. Simple primary and methyl carbocations are not normally seen in solution.

Resonance stabilization of the carbocation can also promote the S_N1 reaction. For example, allyl bromide is a primary halide, but it undergoes the S_N1 reaction about as fast as a secondary halide. The carbocation formed by ionization is resonance-stabilized, with the positive charge spread equally over two carbon atoms.



Vinyl and aryl halides generally do not undergo S_N1 or S_N2 reactions. An S_N1 reaction would require ionization to form a vinyl or aryl cation, either of which is less stable than most alkyl carbocations. An S_N2 reaction would require back-side attack by the nucleophile, which is made impossible by the repulsion of the electrons in the double bond or aromatic ring.

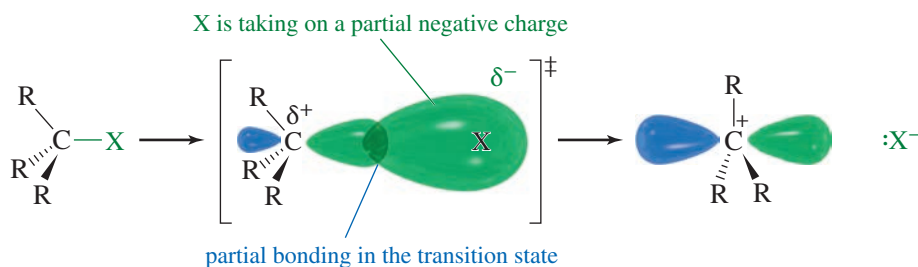


6-13B Leaving-Group Effects

The leaving group is breaking its bond to carbon in the rate-limiting ionization step of the S_N1 mechanism. A highly polarizable leaving group helps stabilize the rate-limiting transition state through partial bonding as it leaves. The leaving group should be a weak base—very stable after it leaves with the pair of electrons that bonded it to carbon.

Figure 6-10 shows the transition state of the ionization step of the S_N1 reaction. Notice how the leaving group is taking on a negative charge while it stabilizes the new carbocation through partial bonding. The leaving group should be stable as it takes on this negative charge, and it should be polarizable to engage in effective partial bonding as it leaves. A good leaving group is just as necessary in the S_N1 reaction as it is in the S_N2 , and similar leaving groups are effective for either reaction. Table 6-4 (page 316) lists some common leaving groups for either reaction.

FIGURE 6-10 In the transition state of the S_N1 ionization, the leaving group is taking on a negative charge. The C—X bond is breaking, and a polarizable leaving group can still maintain substantial overlap.



PROBLEM-SOLVING HINT

Primary cations are rarely formed in solution unless they are resonance-stabilized.

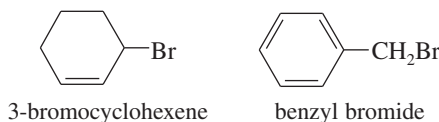
PROBLEM 6-23

Choose the member of each pair that will react faster by the S_N1 mechanism.

- 1-bromopropane or 2-bromopropane
- 2-bromo-2-methylbutane or 2-bromo-3-methylbutane
- n*-propyl bromide or allyl bromide
- 1-bromo-2,2-dimethylpropane or 2-bromopropane
- 2-iodo-2-methylbutane or *tert*-butyl chloride
- 2-bromo-2-methylbutane or ethyl iodide

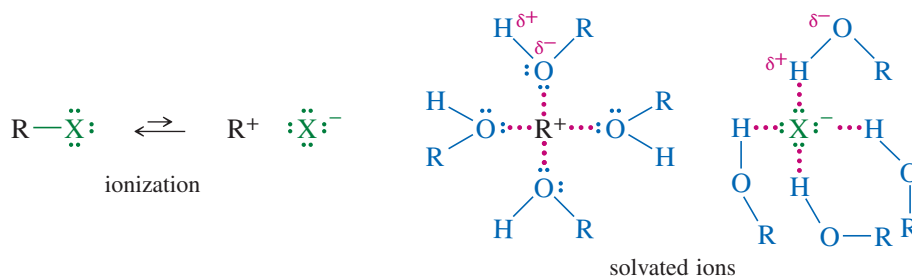
PROBLEM 6-24

3-Bromocyclohexene is a secondary halide, and benzyl bromide is a primary halide. Both halides undergo S_N1 substitution about as fast as most tertiary halides. Use resonance structures to explain this enhanced reactivity.

6-13C Solvent Effects on S_N1 Reactions

The S_N1 reaction goes much more readily in polar solvents that stabilize ions. The rate-limiting step forms two ions, and ionization is taking place in the transition state. Polar solvents solvate these ions by an interaction of the solvent's dipole moment with the charge of the ion. Protic solvents such as alcohols and water are even more effective

solvents because anions form hydrogen bonds with the —OH hydrogen atom, and cations complex with the nonbonding electrons of the —OH oxygen atom.



Ionization of an alkyl halide requires formation and separation of positive and negative charges, similar to what happens when sodium chloride dissolves in water. Therefore, S_N1 reactions require highly polar solvents that strongly solvate ions. One measure of a solvent's ability to solvate ions is its *dielectric constant* (ϵ), a measure of the solvent's polarity. Table 6-6 lists the dielectric constants of some common solvents and the relative ionization rates for *tert*-butyl chloride in these solvents. Note that ionization occurs much faster in highly polar solvents such as water and alcohols. Although most alkyl halides are not soluble in water, they often dissolve in highly polar mixtures of acetone and alcohols with water.

TABLE 6-6 Dielectric Constants (ϵ) and Ionization Rates of *tert*-Butyl Chloride in Common Solvents

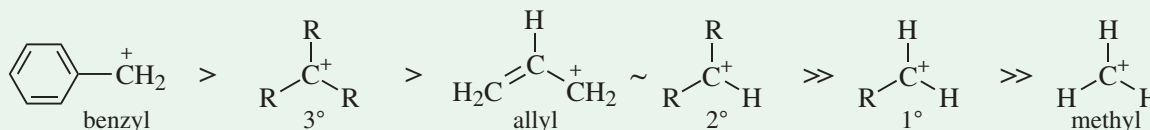
Solvent	ϵ	Relative Rate
water	78	8000
methanol	33	1000
ethanol	24	200
acetone	21	1
diethyl ether	4.3	0.001
hexane	2.0	<0.0001

FOCUS Factors Affecting S_N1 Reaction Rates

The S_N1 reaction follows first order kinetics and the rate depends only on how fast the leaving group leaves in the first step. As a result, nucleophilicity of the nucleophile plays no role in the rate of an S_N1 reaction.

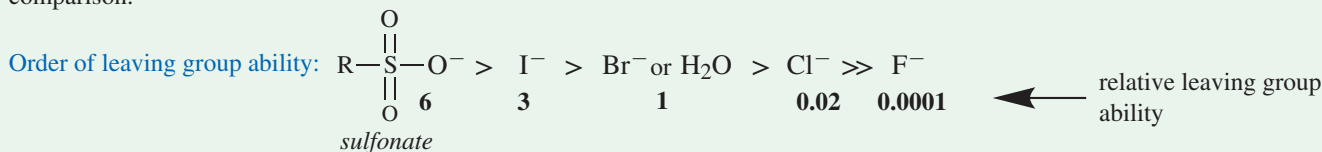
1. Structure. In the first step of an S_N1 reaction, the leaving group leaves, generating a carbocation. Any structural feature that stabilizes the carbocation intermediate will speed the departure of the leaving group, the rate-determining step. The two most common structural factors that stabilize carbocations are greater substitution, and resonance.

Order of carbocation stability:



In practice, only benzylic, 3°, allylic, and 2° alkyl halides undergo the S_N1 reaction. Methyl and 1° halides are too resistant to forming carbocations, and are much more likely to undergo substitution by an S_N2 mechanism.

2. Leaving group. Groups that are highly polarizable (large atoms), and those that are weak bases, are the best leaving groups. This list shows the relative leaving group ability of the more common leaving groups, with Br[−] or H₂O as the standard of comparison.



Strong bases are *never* leaving groups in S_N1 reactions. Strong bases include HO[−], RO[−], H[−], R₂N[−], and R₃C[−].

3. Solvent. Polar solvents, especially protic solvents that can surround and stabilize the leaving group, accelerate S_N1 reactions.

Good polar protic solvents for S_N1: H₂O CH₃OH CH₃CH₂OH CH₃COOH

6-14 Stereochemistry of the S_N1 Reaction

Recall from Section 6-12 that the S_N2 reaction is stereospecific: the nucleophile attacks from the back side of the electrophilic carbon atom, giving inversion of configuration. In contrast, the S_N1 reaction is not stereospecific. In the S_N1 mechanism, the carbocation intermediate is *sp*² hybridized and planar. A nucleophile can attack the carbocation from either face. Figure 6-11 shows the S_N1 solvolysis of a chiral compound, (*S*)-3-bromo-2,3-dimethylpentane, in ethanol. The carbocation is planar and achiral; attack from both faces gives both enantiomers of the product. Such a process, giving both enantiomers of the product (whether or not the two enantiomers are produced in equal amounts), is called **racemization**. The product is either racemic or at least less optically pure than the starting material.

If a nucleophile attacks the carbocation in Figure 6-11 from the front side (the side the leaving group left), the product molecule shows **retention of configuration**. Attack from the back side gives a product molecule showing **inversion of configuration**. Racemization is simply a combination of retention and inversion. When racemization occurs, the product is rarely completely racemic, however; there is often more inversion than retention of configuration. As the leaving group leaves, it partially blocks the front side of the carbocation. The back side is unhindered, so attack is more likely there.

Figure 6-12 shows a cyclic case where one of the faces of a cyclopentane ring has been “labeled” by a deuterium atom. Deuterium has the same size and shape as hydrogen, and it undergoes the same reactions. It distinguishes between the two faces of the ring: the bromine atom is *cis* to the deuterium in the reactant, so the nucleophile is *cis* to the deuterium in the retention product. The nucleophile is *trans* to the deuterium in the inversion product. The product mixture contains both *cis* and *trans* isomers, with the *trans* isomer slightly favored because the leaving group hinders approach of the nucleophilic solvent from the front side.

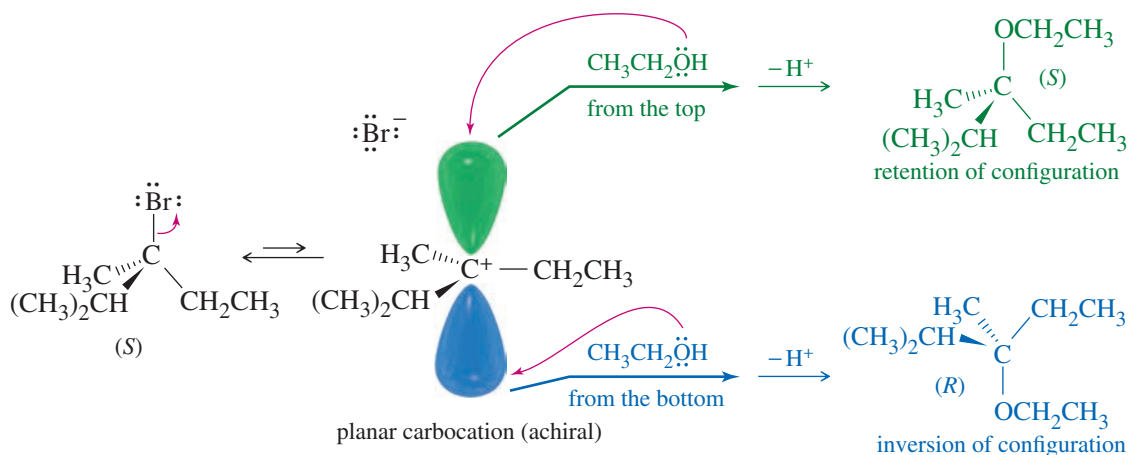


FIGURE 6-11 Racemization. An asymmetric carbon atom undergoes racemization when it ionizes to a planar, achiral carbocation. A nucleophile can attack the carbocation from either face, giving either enantiomer of the product.

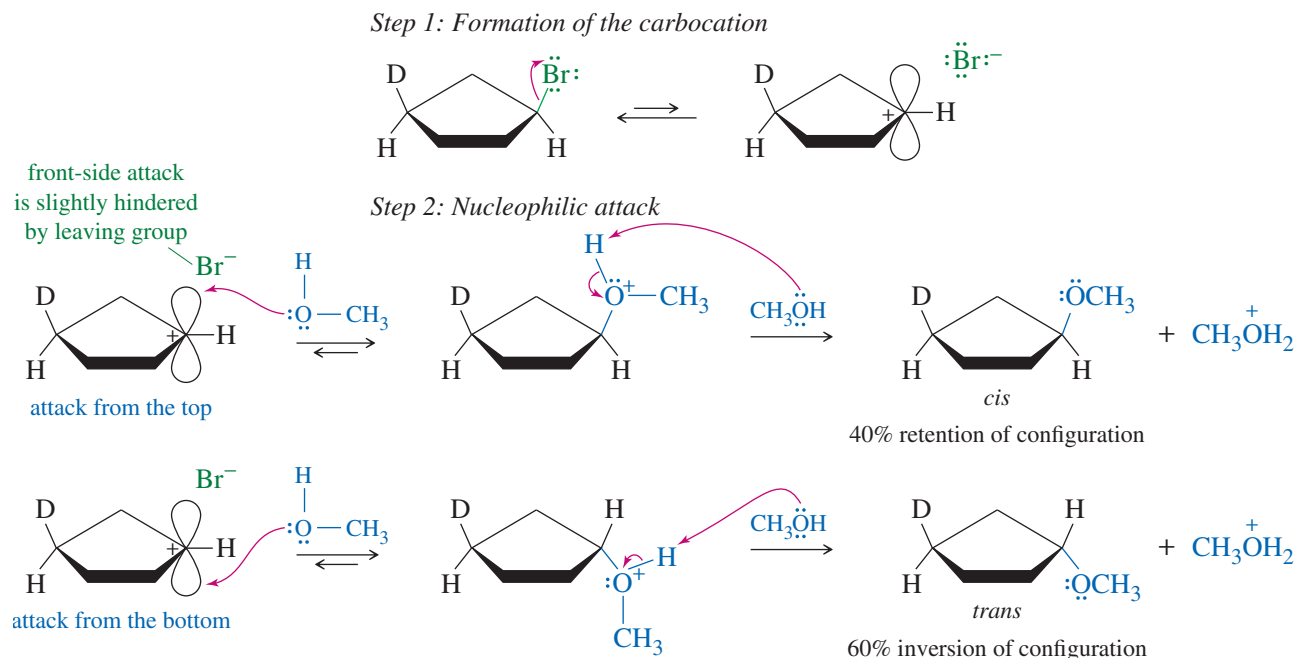
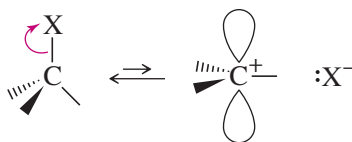


FIGURE 6-12 In the S_N1 reaction of *cis*-1-bromo-3-deuteriocyclopentane with methanol, the carbocation can be attacked from either face. Because the leaving group (bromide) partially blocks the front side as it leaves, back-side attack (inversion of configuration) is slightly favored.

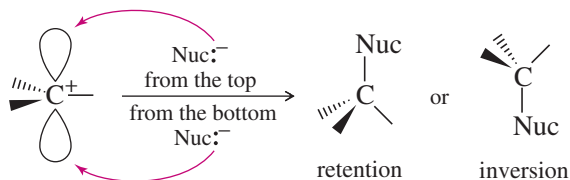
MECHANISM 6-5 Racemization in the S_N1 Reaction

The S_N1 reaction involves ionization to a flat carbocation, which can be attacked from either side.

Step 1: Ionization of a tetrahedral carbon gives a flat carbocation.



Step 2: A nucleophile may attack either side of the carbocation.



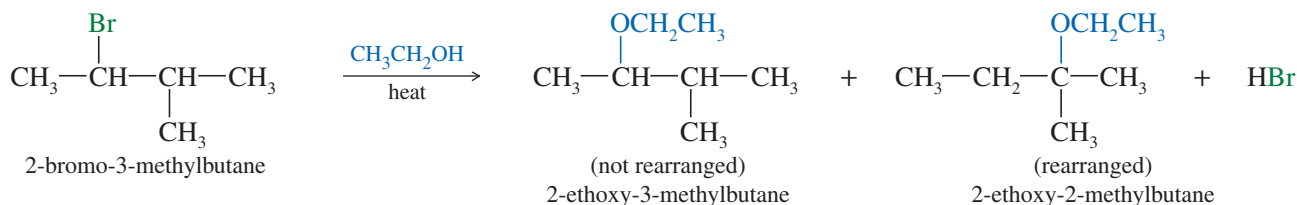
Racemization: Formation of products showing both retention and inversion of configuration.

These two products may be different if the carbon atom is stereogenic.

6-15 Rearrangements in S_N1 Reactions

Carbocations frequently undergo structural changes, called **rearrangements**, to form more stable ions. A rearrangement may occur after a carbocation has formed, or it may occur as the leaving group is leaving. Rearrangements are not seen in S_N2 reactions, where no carbocation is formed and the one-step mechanism allows no opportunity for rearrangement.

An example of a reaction with rearrangement is the S_N1 reaction of 2-bromo-3-methylbutane in boiling ethanol. The product is a mixture of 2-ethoxy-3-methylbutane (not rearranged) and 2-ethoxy-2-methylbutane (rearranged).



PROBLEM-SOLVING HINT

A carbocation usually rearranges just once. Most rearrangements convert the carbocation to a more stable carbocation, typically
 $2^\circ \longrightarrow 3^\circ$
 or $2^\circ \longrightarrow$ resonance-stabilized
 or $3^\circ \longrightarrow$ resonance-stabilized.

PROBLEM 6-25

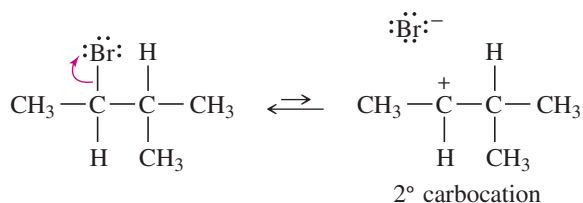
Give the S_N1 mechanism for the formation of 2-ethoxy-3-methylbutane, the unrearranged product in this reaction.

The rearranged product, 2-ethoxy-2-methylbutane, results from a **hydride shift**, the movement of a hydrogen atom with its bonding pair of electrons. A hydride shift is represented by the symbol $\sim\text{H}$. In this case, the hydride shift converts the initially formed secondary carbocation to a more stable tertiary carbocation. Attack by the solvent gives the rearranged product.

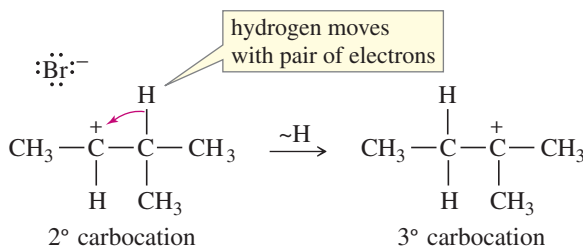
MECHANISM 6-6 Hydride Shift in an S_N1 Reaction

Carbocations often rearrange to form more stable carbocations. This may occur when a hydrogen atom moves with its bonding pair of electrons. Formally, this is the movement of a hydride ion (H^-), although no actual free hydride ion is involved.

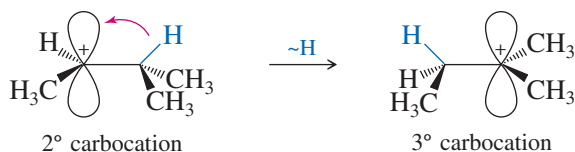
Step 1: Unimolecular ionization gives a carbocation.



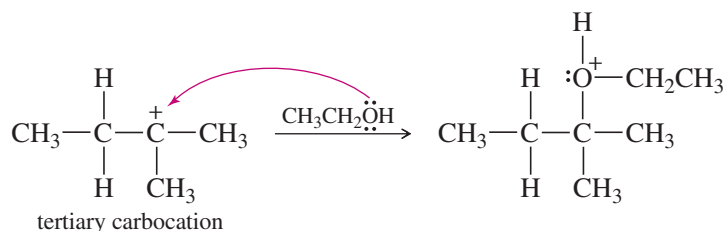
Step 2: A hydride shift forms a more stable carbocation.



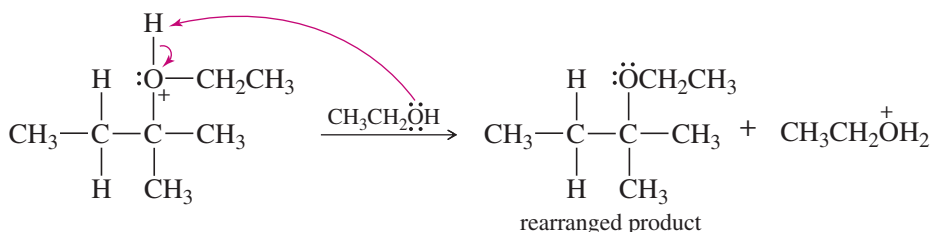
This rearrangement involves movement of a hydrogen atom with its bonding pair of electrons over to the empty *p* orbital of the carbocation. In three dimensions, the rearrangement looks like this:



Step 3: Solvent (a weak nucleophile) attacks the rearranged carbocation.



Step 4: Deprotonation gives the rearranged product.



When neopentyl bromide is boiled in ethanol, it gives *only* a rearranged substitution product. This product results from a **methyl shift** (represented by the symbol $\sim\text{CH}_3$), the migration of a methyl group together with its pair of electrons. Without rearrangement, ionization of neopentyl bromide would give a very unstable primary carbocation.



PROBLEM-SOLVING HINT

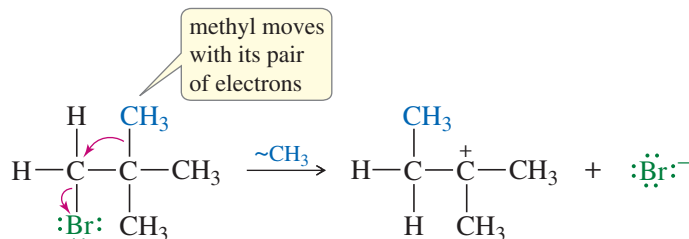
Primary halides and methyl halides rarely ionize to carbocations in solution. If a primary halide ionizes, it will likely ionize with rearrangement.

The methyl shift occurs *while* bromide ion is leaving, so that only the more stable tertiary carbocation is formed.

MECHANISM 6-7 Methyl Shift in an S_N1 Reaction

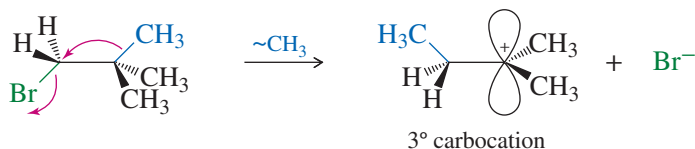
An alkyl group can rearrange to make a carbocation more stable.

Step 1: Ionization occurs with a methyl shift.

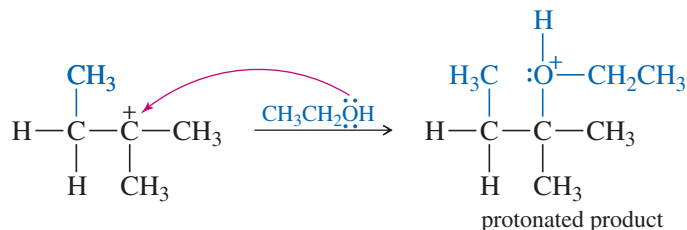


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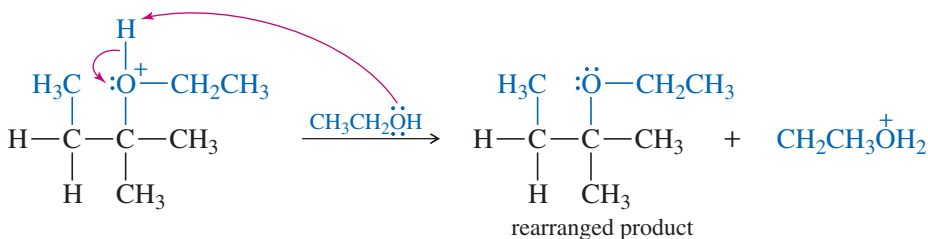
In three dimensions,



Step 2: Attack by ethanol gives a protonated version of the rearranged product.



Step 3: Deprotonation gives the rearranged product.



Because rearrangement is required for ionization of this primary halide, only rearranged products are observed.

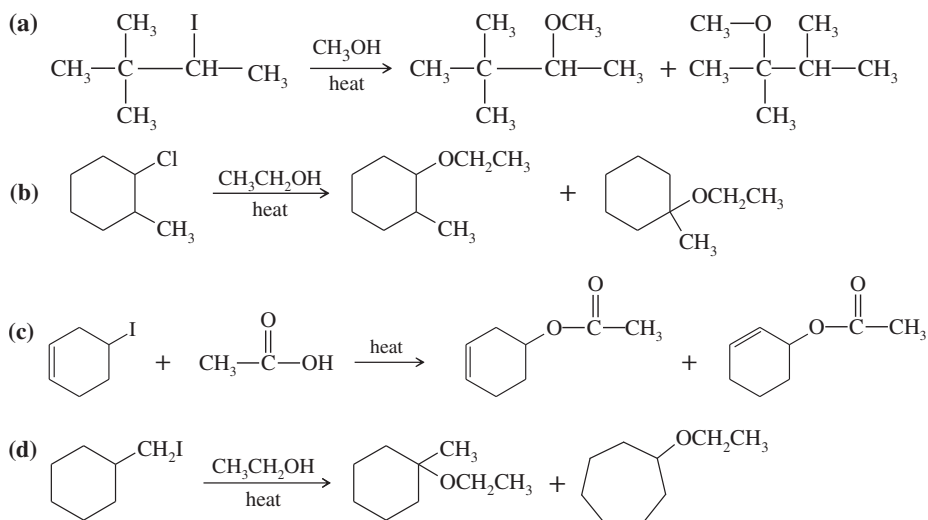
In general, we should expect rearrangements in reactions involving carbocations whenever a hydride shift or an **alkyl shift** can form a more stable carbocation. Most rearrangements convert 2° (or incipient 1°) carbocations to 3° or resonance-stabilized carbocations.

PROBLEM-SOLVING HINT

Most rearrangements convert 2° (or incipient 1°) carbocations to 3° or resonance-stabilized carbocations.

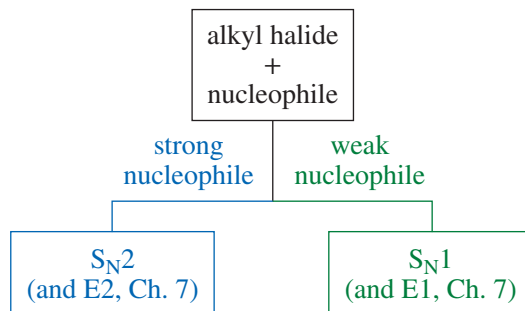
PROBLEM 6-26

Propose a mechanism involving a hydride shift or an alkyl shift for each solvolysis reaction. Explain how each rearrangement forms a more stable intermediate.



6-16 Comparison of S_N1 and S_N2 Reactions

Let's compare what we know about the S_N1 and S_N2 reactions, then organize this material into a brief table.



Effect of the Nucleophile The nucleophile takes part in the slow step (the only step) of the S_N2 reaction but not in the slow step of the S_N1 . Therefore, a strong nucleophile promotes the S_N2 but not the S_N1 . Weak nucleophiles fail to promote the S_N2 reaction; therefore, reactions with weak nucleophiles often go by the S_N1 mechanism if the substrate is secondary or tertiary.

S_N1 : Nucleophile strength is unimportant (usually weak).
 S_N2 : Strong nucleophiles are required.

Effect of the Substrate The structure of the substrate (the alkyl halide) is an important factor in determining which of these substitution mechanisms might operate. Most methyl halides and primary halides are poor substrates for S_N1 substitutions because they cannot easily ionize to high-energy methyl and primary carbocations. They are relatively unhindered, however, so they make good S_N2 substrates.

Tertiary halides are too hindered to undergo S_N2 displacement, but they can ionize to form tertiary carbocations. Tertiary halides undergo substitution exclusively through the S_N1 mechanism. Secondary halides can undergo substitution by either mechanism, depending on the conditions.

S_N1 substrates: $3^\circ > 2^\circ$ (1° and CH_3X are unlikely)
 S_N2 substrates: $\text{CH}_3\text{X} > 1^\circ > 2^\circ$ (3° is unsuitable)

If silver nitrate (AgNO_3) is added to an alkyl halide in a good ionizing solvent, the silver ion removes the halide ion to give a carbocation. This technique can force some unlikely ionizations, often giving interesting rearrangements (see Problem 6-29).

Effect of the Solvent The slow step of the S_N1 reaction involves formation of two ions. Solvation of these ions is crucial to stabilizing them and lowering the activation energy for their formation. Very polar ionizing solvents such as water and alcohols are needed for the S_N1 . The solvent may be heated to reflux (boiling) to provide the energy needed for ionization.

Less charge separation is generated in the transition state of the S_N2 reaction. Strong solvation may weaken the strength of the nucleophile because of the energy needed to strip off the solvent molecules. Thus, the S_N2 reaction often goes faster in less polar solvents if the nucleophile will dissolve. Polar aprotic solvents may enhance the strength of weak nucleophiles.

S_N1 : Good ionizing solvent required.
 S_N2 : May go faster in a less polar solvent.

Kinetics The rate of the S_N1 reaction is proportional to the concentration of the alkyl halide but not the concentration of the nucleophile. It follows a first-order rate equation.

The rate of the S_N2 reaction is proportional to the concentrations of both the alkyl halide $[R-X]$ and the nucleophile $[Nuc:]^-]$. It follows a second-order rate equation.

$$\begin{aligned} S_N1 \text{ rate} &= k_t[R-X] \\ S_N2 \text{ rate} &= k_t[R-X][Nuc:]^-] \end{aligned}$$

Stereochemistry The S_N1 reaction involves a flat carbocation intermediate that can be attacked from either face. Therefore, the S_N1 usually gives a mixture of inversion and retention of configuration.

The S_N2 reaction takes place through a back-side attack, which inverts the stereochemistry of the carbon atom. Complete inversion of configuration is the result.

$$\begin{aligned} S_N1 \text{ stereochemistry:} & \quad \text{Mixture of retention and inversion; racemization.} \\ S_N2 \text{ stereochemistry:} & \quad \text{Complete inversion.} \end{aligned}$$

Rearrangements The S_N1 reaction involves a carbocation intermediate. This intermediate can rearrange, usually by a hydride shift or an alkyl shift, to give a more stable carbocation.

The S_N2 reaction takes place in one step with no intermediates. No rearrangement is possible in the S_N2 reaction.

$$\begin{aligned} S_N1: & \quad \text{Rearrangements are common.} \\ S_N2: & \quad \text{Rearrangements are impossible.} \end{aligned}$$

SUMMARY Nucleophilic Substitutions

	S_N1	S_N2
Promoting factors		
nucleophile	weak nucleophiles are OK	strong nucleophile needed
substrate (RX)	$3^\circ > 2^\circ$	$CH_3X > 1^\circ > 2^\circ$
solvent	good ionizing solvent needed	wide variety of solvents
leaving group	good one required	good one required
other	$AgNO_3$ forces ionization	
Characteristics		
kinetics	first order, $k_t[RX]$	Second order, $k_t[RX][Nuc:]^-]$
stereochemistry	mixture of inversion and retention	complete inversion
rearrangements	common	impossible
	S_N1 conditions (weak nucleophile)	S_N2 conditions (strong nucleophile)
methyl halides CH_3X	No reaction. A methyl cation is too unstable.	S_N2 is unhindered and favored.
primary halides RCH_2X	Rarely any reaction unless it can form a resonance-stabilized cation.	S_N2 is favored unless the R group is exceptionally bulky.
secondary halides R_2CHX	S_N1 , often with rearrangement, can occur in a good solvent.	S_N2 can occur unless the alkyl groups or the nucleophile are bulky.
tertiary halides R_3CX	S_N1 occurs readily in a good solvent.	S_N2 cannot occur because of steric hindrance.

PROBLEM 6-27

For each reaction, give the expected substitution product, and predict whether the mechanism will be predominantly first order (S_N1) or second order (S_N2).

- (a) 2-chloro-2-methylbutane + CH₃COOH
 (b) isobutyl bromide + sodium methoxide
 (c) 1-iodo-1-methylcyclohexane + ethanol
 (d) cyclohexyl bromide + methanol
 (e) cyclohexyl bromide + sodium ethoxide

PROBLEM 6-28

Under certain conditions, when (*R*)-2-bromobutane is heated with water, the S_N1 substitution proceeds twice as fast as the S_N2. Calculate the e.e. and the specific rotation expected for the product. The specific rotation of (*R*)-butan-2-ol is −13.5°. Assume that the S_N1 gives equal amounts of the two enantiomers.

PROBLEM-SOLVING HINT

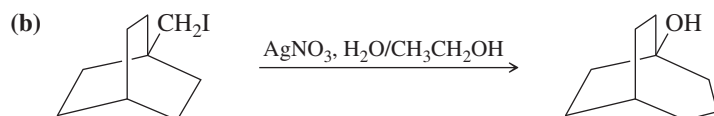
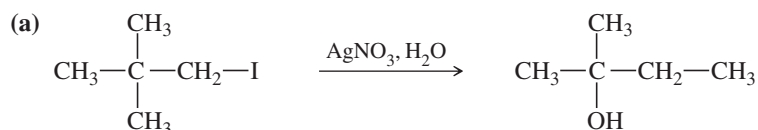
The strength of the nucleophile (or base) usually determines the order of the reaction. Strong nucleophiles encourage bimolecular, second-order reactions, and weak nucleophiles more commonly react by unimolecular, first-order mechanisms. Also, S_N2 is unlikely with 3° halides, and S_N1 is unlikely with 1° halides unless they are resonance-stabilized.

PROBLEM 6-29

A reluctant first-order substrate can be forced to ionize by adding some silver nitrate (one of the few soluble silver salts) to the reaction. Silver ion reacts with the halogen to form a silver halide (a highly exothermic reaction), generating the cation of the alkyl group.



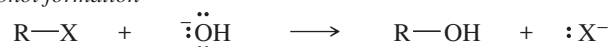
Give mechanisms for the following silver-promoted rearrangements.



The Rest of the Story Elimination reactions often accompany nucleophilic substitutions of alkyl halides and other compounds containing good leaving groups. Under conditions that favor them, elimination reactions often predominate over substitutions. In Chapter 6, we have concentrated on the factors that favor S_N1 and S_N2 substitutions. In Chapter 7, we will cover the factors that favor eliminations. At that point, we can begin to predict the products and mechanisms of many reactions that are subject to competition between substitutions and eliminations.

SUMMARY Reactions of Alkyl Halides

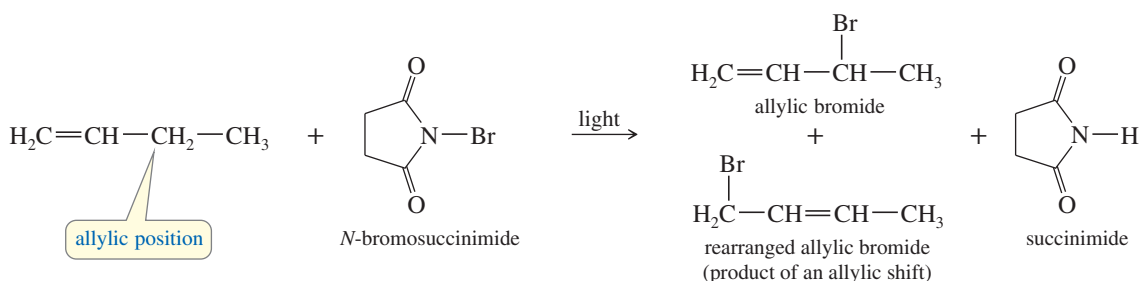
Some of these reactions have not yet been covered, but they are included here for completeness and for later reference. The reactions that we have not yet covered are shown in the summary with a gray background. Notice the section numbers, indicating where each reaction is covered.

1. Nucleophilic substitutions (Section 6-9)**a. Alcohol formation**

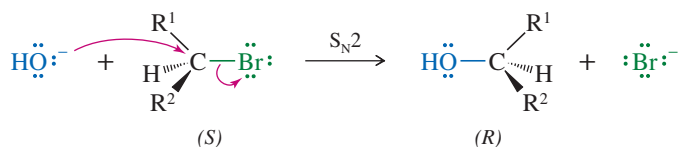
(continued)

Essential Terms

acid	A species that can donate a proton. (p. 107)
acidity:	(acid strength) The thermodynamic reactivity of an acid, expressed quantitatively by the acid-dissociation constant K_a . (p. 109)
Lewis acid:	(electrophile) A species that can accept an electron pair from a nucleophile, forming a bond. (pp. 126, 317)
alkyl halide	(haloalkane) A derivative of an alkane in which one (or more) of the hydrogen atoms has been replaced by a halogen. (p. 292)
alkyl shift	(symbolized ~R) Movement of an alkyl group with a pair of electrons from one atom (usually carbon) to another. Alkyl shifts are examples of rearrangements that convert carbocations into more stable carbocations. (p. 330)
allylic	The saturated position adjacent to a carbon-carbon double bond. (p. 301)
allylic halogenation	Substitution of a halogen for a hydrogen at the allylic position. (p. 301)

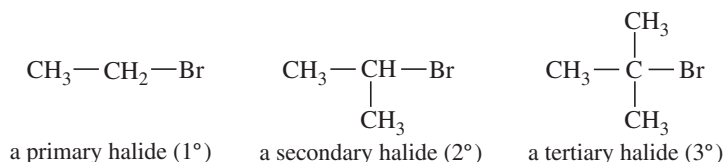


allylic shift	A rearrangement that results from reaction at either end of a resonance-stabilized allylic intermediate. (p. 302)
aprotic solvent	A solvent that has no acidic protons; a solvent with no O—H or N—H groups. (p. 313)
aryl halide	An aromatic compound (benzene derivative) in which a halogen is bonded to one of the carbon atoms of the aromatic ring. (p. 292)
base	An electron-rich species that can abstract a proton. (pp. 107, 311)
basicity:	(base strength) The thermodynamic reactivity of a base, expressed quantitatively by the base-dissociation constant K_b . (pp. 110, 311)
Lewis base:	(nucleophile) An electron-rich species that can donate a pair of electrons to form a bond. (p. 126)
concerted reaction	A reaction in which the breaking of bonds and the formation of new bonds occur at the same time (in one step). (p. 307)
dehydrohalogenation	An elimination in which the two atoms lost are a hydrogen atom and a halogen atom. (p. 305)
electrophile	(Lewis acid) A species that can accept an electron pair from a nucleophile, forming a bond. (p. 126)
electrophilicity:	(electrophile strength) The kinetic reactivity of an electrophile. (p. 317)
elimination	A reaction that involves the loss of two atoms or groups from the substrate, usually resulting in the formation of a pi bond. (p. 305)
freons	A generic name for a group of chlorofluorocarbons used as refrigerants, propellants, and solvents. Freon-12® is CF_2Cl_2 , and Freon-22® is CHClF_2 . (p. 296)
geminal dihalide	A dihalide with both halogens on the same carbon atom. (p. 294) $\text{CH}_3-\text{CH}_2-\text{CBr}_2-\text{CH}_3$ a geminal dibromide
haloalkane	(alkyl halide) A derivative of an alkane in which one (or more) of the hydrogen atoms has been replaced by a halogen. (p. 293)
halogen exchange reaction	A substitution where one halogen atom replaces another; commonly used to form fluorides and iodides. (p. 309)
hydride shift	(symbolized ~H) Movement of a hydrogen atom with a pair of electrons from one atom (usually carbon) to another. Hydride shifts are examples of rearrangements that convert carbocations into more stable carbocations. (p. 328)
hydroxylic solvent	A solvent containing OH groups (the most common type of protic solvents). (p. 313)
inversion of configuration	(see also Walden inversion) A process in which the groups around an asymmetric carbon atom are changed to the opposite spatial configuration, usually as a result of back-side attack . (p. 318)

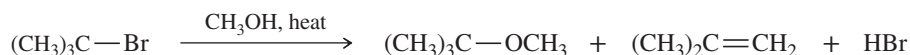


The $\text{S}_\text{N}2$ reaction goes with *inversion of configuration*.

leaving group	The atom or group of atoms that departs during a substitution or elimination. The leaving group can be charged or uncharged, but it leaves with the pair of electrons that originally bonded the group to the remainder of the molecule. (p. 305)
Lewis acid	See electrophile . (pp. 126, 317)
Lewis base	See nucleophile . (pp. 126, 306)
methyl shift	(symbolized $\sim\text{CH}_3$) Rearrangement of a methyl group with a pair of electrons from one atom (usually carbon) to another. A methyl shift (or any alkyl shift) in a carbocation generally results in a more stable carbocation. (p. 329)
NBS (N-bromosuccinimide)	A reagent that is often used as the bromine source in allylic bromination. NBS provides a consistently low concentration of Br_2 as the reaction proceeds because it combines with the HBr byproduct to regenerate Br_2 . (p. 302)
nucleophile	(Lewis base) An electron-rich species that can donate a pair of electrons to form a bond. (pp. 126, 306)
nucleophilicity:	(nucleophile strength) The kinetic reactivity of a nucleophile; a measure of the rate of substitution in a reaction with a standard substrate. (p. 311)
nucleophilic substitution	A reaction where a nucleophile replaces another group or atom (the leaving group) in a molecule. (p. 305)
organic synthesis	The preparation of desired organic compounds from readily available starting materials. (p. 295)
polarizable	Having electrons that are easily displaced toward a positive charge. Polarizable atoms can begin to form a bond at a relatively long distance. (p. 316)
primary halide, secondary halide, tertiary halide	These terms specify the substitution of the halogen-bearing carbon atom (sometimes called the head carbon). If the head carbon is bonded to one other carbon, it is primary ; if it is bonded to two carbons, it is secondary ; and if bonded to three carbons, it is tertiary . (p. 294)



protic solvent	A solvent containing acidic protons, usually $\text{O}-\text{H}$ or $\text{N}-\text{H}$ groups. (p. 313)
racemization	The loss of optical activity that occurs when a reaction shows neither clean retention of configuration nor clean inversion of configuration. (p. 326)
reagent	The compound that serves as the attacking species in a reaction. (p. 295)
rearrangement	A reaction involving a change in the bonding sequence within a molecule. Rearrangements are common in reactions such as the $\text{S}_\text{N}1$ and $\text{E}1$ involving carbocation intermediates. (p. 328)
retention of configuration	Formation of a product with the same configuration as the reactant. In a nucleophilic substitution, retention of configuration occurs when the nucleophile assumes the same stereochemical position in the product as the leaving group occupied in the reactant. (p. 326)
solvolysis	A nucleophilic substitution or elimination where the solvent serves as the attacking reagent. Solvolysis literally means "cleavage by the solvent." (p. 320)



stereospecific reaction	A reaction in which different stereoisomers of the starting material react to give different stereoisomers of the product. (p. 319)
steric hindrance	Interference by bulky groups that slow a reaction or prevent it from occurring. (p. 317)
substitution	(displacement) A reaction in which an attacking species (nucleophile, electrophile, or free radical) replaces another group. (p. 321)
$\text{S}_\text{N}2$ reaction:	(Substitution, Nucleophilic, bimolecular) The concerted displacement of one nucleophile by another on an sp^3 hybrid carbon atom. (p. 322)

S_N1 reaction:	(Substitution, Nucleophilic, unimolecular) A two-step interchange of nucleophiles, with bond breaking preceding bond formation. The first step is ionization to form a carbocation. The second step is the reaction of the carbocation with a nucleophile. (p. 321)
substrate	The compound that is attacked by the reagent. (p. 306)
transition state	In each individual step of a reaction, the state of highest energy between reactants and products. The transition state is a relative maximum (high point) on the reaction-energy diagram. (p. 307)
vicinal dihalide	A dihalide with the halogens on adjacent carbon atoms. (p. 294)
$\text{CH}_3\text{—CHBr—CHBr—CH}_3$ a vicinal dibromide $\text{CH}_3\text{—CH=}\overset{\text{Br}}{\underset{ }{\text{C}}}\text{—CH}_3$ a vinyl bromide	
vinyl halide	A derivative of an alkene in which one (or more) of the hydrogen atoms on the double-bonded carbon atoms has been replaced by a halogen. (p. 292)
Walden inversion	(see also inversion of configuration) A step in a reaction sequence in which an asymmetric carbon atom undergoes inversion of configuration. (p. 319)

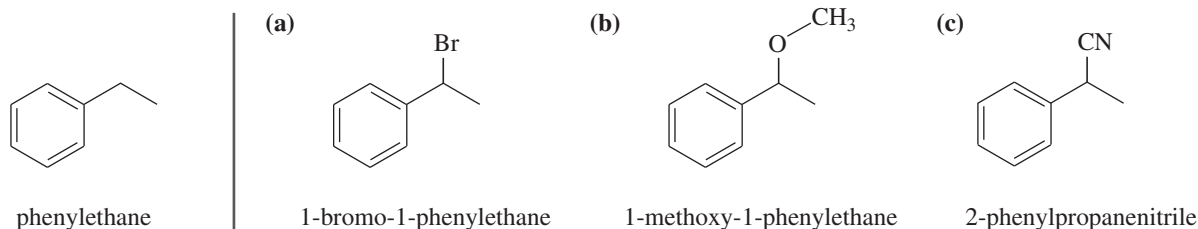
Essential Problem-Solving Skills in Chapter 6

Each skill is followed by problem numbers exemplifying that particular skill.

- | | |
|---|---|
| 1 Correctly name alkyl halides; summarize their physical properties; and identify them as 1°, 2°, or 3°. | Problems 6-31, 32, and 47 |
| 2 Show how free-radical halogenation might be used for the synthesis of some alkyl halides, especially for making allylic and benzylic alkyl halides. | Problems 6-30, 46, 51, 57, 58, and 59 |
| 3 Predict the products of S _N 1 and S _N 2 reactions, including stereochemistry. | Problems 6-35, 40, 41, 44, 45, 46, and 52 |
| 4 Draw the mechanisms and reaction-energy diagrams of S _N 1 and S _N 2 reactions. | Problems 6-40, 41, 48, 49, 52, 53, 54, 55, and 56 |
| 5 Predict and explain the relative stabilities and rearrangements of cations involved in first-order reactions. | Problems 6-40, 41, 42, 43, and 54 |
| 6 Predict which substitutions will be faster, based on differences in substrate, nucleophile, leaving group, and solvent. | Problems 6-33, 34, 36, 37, 38, and 39 |
| 7 Given a set of reaction conditions, predict whether the reaction will be unimolecular (first-order) or bimolecular (second-order), and predict which product(s) are most likely. | Problems 6-37, 38, 54, and 56 |
| 8 Show how substitutions of alkyl halides might be used to synthesize other types of compounds. | Problems 6-30, 35, 36, 39, 40, 45, 46, 53, and 54 |

Study Problems

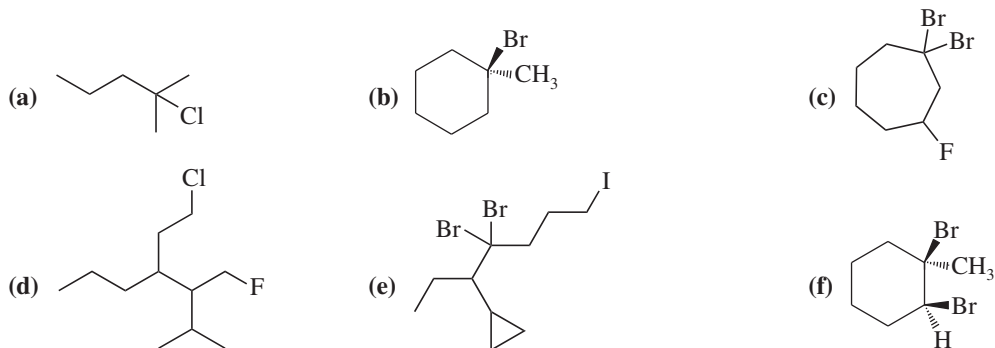
- 6-30** Show how you would convert (in one or two steps) phenylethane to the three products shown below. In each case, explain what unwanted reactions might produce undesirable impurities in the product.



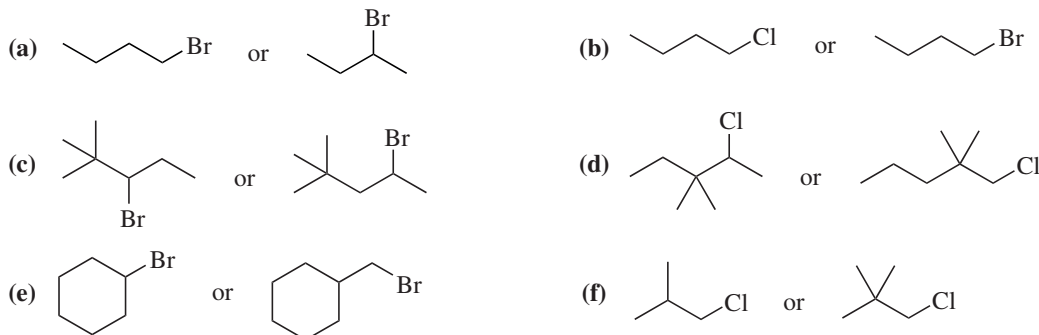
- 6-31** Draw the structures of the following compounds.

- | | | |
|--------------------------------------|--|--|
| (a) <i>sec</i> -butyl bromide | (b) isobutyl chloride | (c) 1,2-chloro-3-methylpentane |
| (d) 2,2,2-tribromoethanol | (e) <i>trans</i> -1-bromo-2-methylcyclohexane | (f) methylene bromide |
| (g) iodoform | (h) 1-bromo-1-isopropylcyclopentane | (i) <i>tert</i> -pentyl bromide |

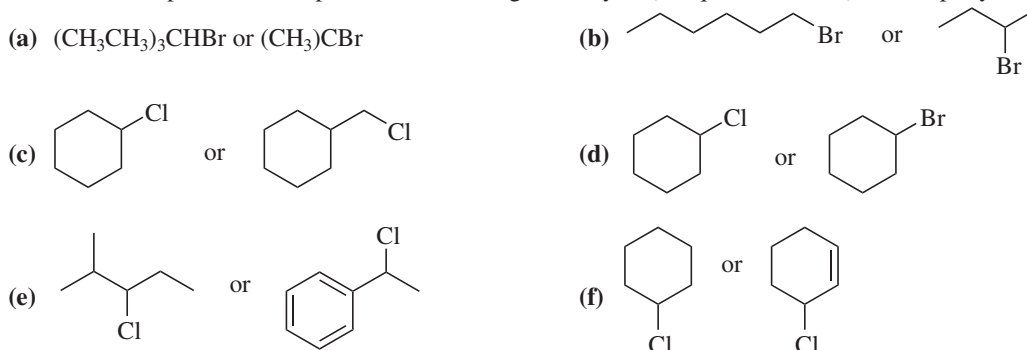
6-32 Give systematic (IUPAC) names for the following compounds.



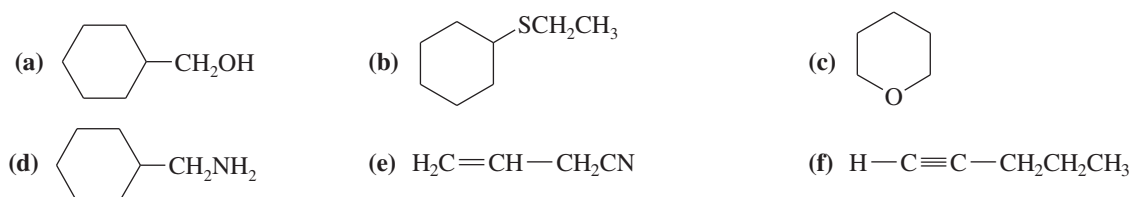
6-33 Predict the compound in each pair that will undergo the S_N2 reaction faster.



6-34 Predict the compound in each pair that will undergo solvolysis (in aqueous ethanol) more rapidly.

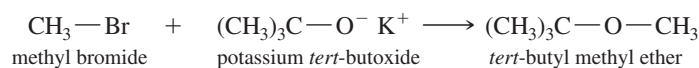


6-35 Show how each compound might be synthesized by the S_N2 displacement of an alkyl halide.



6-36 Give two syntheses for $(\text{CH}_3)_2\text{CH}-\text{O}-\text{CH}_3$, and explain which synthesis is better.

6-37 When methyl bromide is added to potassium *tert*-butoxide, the product is *tert*-butyl methyl ether.

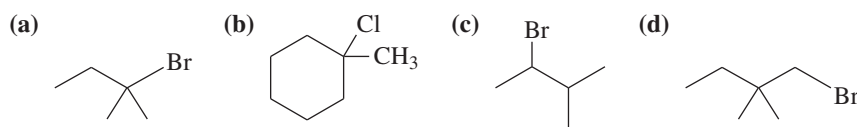


- (a) What happens to the reaction rate if the concentration of methyl bromide is doubled?
 (b) What happens to the rate if the concentration of potassium *tert*-butoxide is tripled and the concentration of methyl bromide is doubled?
 (c) What happens to the rate if the temperature is raised?

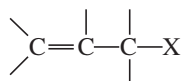
6-38 When *tert*-butyl chloride is heated with an equal amount of ethanol in an inert solvent, one of the products is ethyl *tert*-butyl ether.

- (a) What happens to the reaction rate if the concentration of ethanol is doubled?
 (b) What happens to the rate if the concentration of *tert*-butyl chloride is tripled and the concentration of ethanol is doubled?
 (c) What happens to the rate if the temperature is raised?

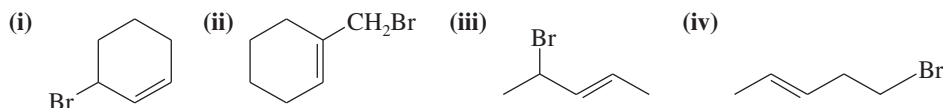
- 6-39** Chlorocyclohexane reacts with sodium cyanide (NaCN) in ethanol to give cyanocyclohexane. The rate of formation of cyanocyclohexane increases when a small amount of sodium iodide is added to the solution. Explain this acceleration in the rate.
- 6-40** Give the substitution products expected from solvolysis of each compound by heating in ethanol.



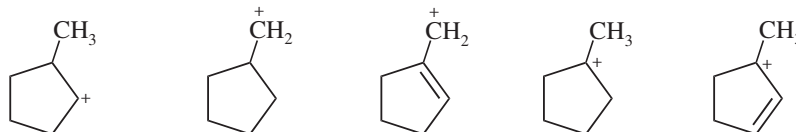
- 6-41** Allylic halides have the structure



- (a) Show how the first-order ionization of an allylic halide leads to a resonance-stabilized cation.
 (b) Draw the resonance structures of the allylic cations formed by ionization of the following halides.
 (c) Show the products expected from $\text{S}_{\text{N}}1$ solvolysis of these halides in ethanol.

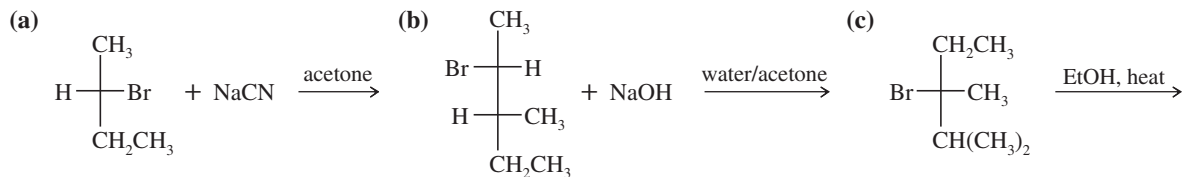


- 6-42** List the following carbocations in decreasing order of their stability.

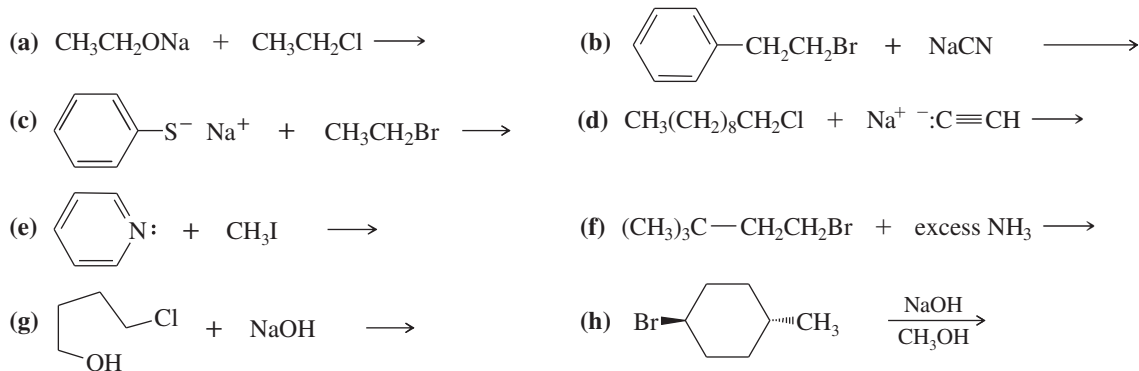


- 6-43** Two of the carbocations in Problem 6-42 are prone to rearrangement. Show how they might rearrange to more stable carbocations.

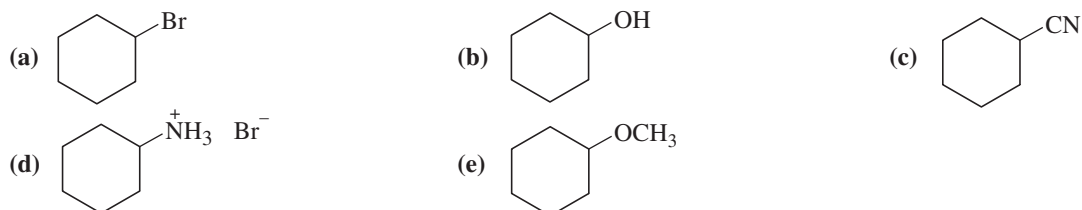
- 6-44** Draw perspective structures or Fischer projections for the substitution products of the following reactions.



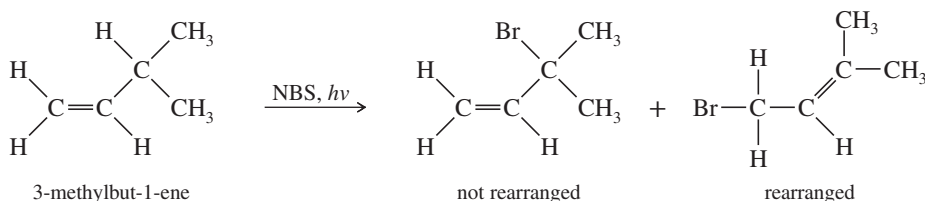
- 6-45** Predict the products of the following $\text{S}_{\text{N}}2$ reactions.



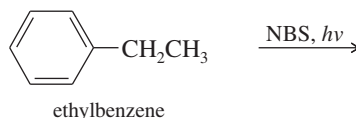
- 6-46** Using cyclohexane as one of your starting materials, show how you would synthesize the following compounds.



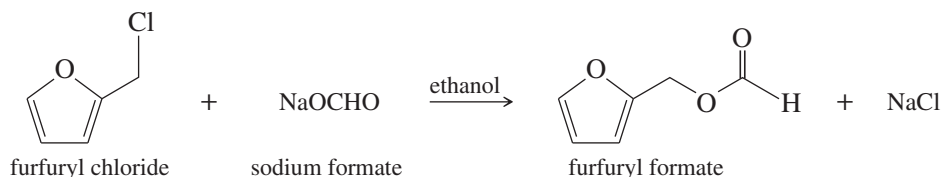
- 6-47** Strawberry growers have used large quantities of methyl bromide (b.p. 4°C) to sterilize the soil before planting their crops. Like some of the freons, methyl bromide can diffuse up into the stratosphere, where it damages the protective ozone layer. Agricultural chemists have suggested using methyl iodide (b.p. 43°C) as a replacement for methyl bromide. Why is methyl iodide likely to be more toxic to agricultural pests (and people) than methyl bromide? Why is methyl iodide less likely to reach the stratosphere than methyl bromide?
- 6-48** A solution of pure (*S*)-2-iodobutane ($[\alpha] = +15.90^{\circ}$) in acetone is allowed to react with radioactive iodide, $^{131}\text{I}^{-}$, until 1.0% of the iodobutane contains radioactive iodine. The specific rotation of this recovered iodobutane is found to be $+15.58^{\circ}$.
 (a) Determine the percentages of (*R*)- and (*S*)-2-iodobutane in the product mixture.
 (b) What does this result suggest about the mechanism of the reaction of 2-iodobutane with iodide ion?
- 6-49** (a) Optically active 2-bromobutane undergoes racemization on treatment with a solution of KBr. Give a mechanism for this racemization.
 (b) In contrast, optically active butan-2-ol does not racemize on treatment with a solution of KOH. Explain why a reaction like that in part (a) does not occur.
 (c) Optically active butan-2-ol racemizes in dilute acid. Propose a mechanism for this racemization.
- 6-50** Give a mechanism to explain the two products formed in the following reaction.



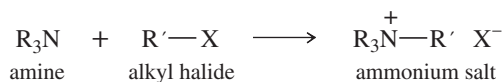
- 6-51** Predict the major product of the following reaction, and give a mechanism to support your prediction.



- 6-52** Because the $\text{S}_{\text{N}}1$ reaction goes through a flat carbocation, we might expect an optically active starting material to give a completely racemized product. In most cases, however, $\text{S}_{\text{N}}1$ reactions actually give more of the *inversion* product. In general, as the stability of the carbocation increases, the excess inversion product decreases. Extremely stable carbocations give completely racemic products. Explain these observations.
- 6-53** Triethyloxonium tetrafluoroborate, $(\text{CH}_3\text{CH}_2)_3\text{O}^+ \text{BF}_4^-$, is a solid with melting point $91\text{--}92^{\circ}\text{C}$. Show how this reagent can transfer an ethyl group to a nucleophile (Nuc^-) in an $\text{S}_{\text{N}}2$ reaction. What is the leaving group? Why might this reagent be preferred to using an ethyl halide? (Consult Table 6-2.)
- 6-54** Furfuryl chloride can undergo substitution by both $\text{S}_{\text{N}}2$ and $\text{S}_{\text{N}}1$ mechanisms. Because it is a 1° alkyl halide, we expect $\text{S}_{\text{N}}2$ but not $\text{S}_{\text{N}}1$ reactions. Draw a mechanism for the $\text{S}_{\text{N}}1$ reaction shown below, paying careful attention to the structure of the intermediate. How can this primary halide undergo $\text{S}_{\text{N}}1$ reactions?

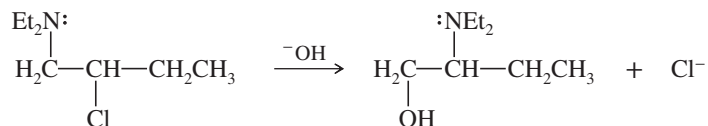


- 6-55** The reaction of an amine with an alkyl halide gives an ammonium salt.

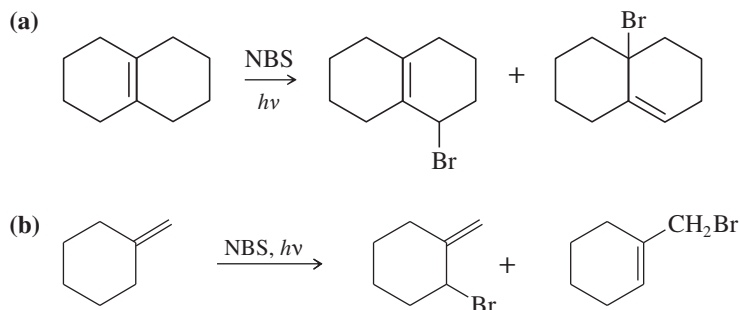


The rate of this $\text{S}_{\text{N}}2$ reaction is sensitive to the polarity of the solvent. Draw an energy diagram for this reaction in a nonpolar solvent and another in a polar solvent. Consider the nature of the transition state, and explain why this reaction should be sensitive to the polarity of the solvent. Predict whether it will be faster or slower in a more polar solvent.

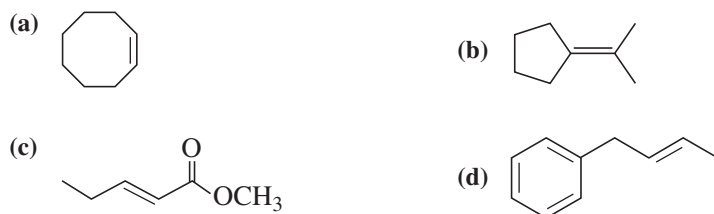
- 6-56** The following reaction takes place under second-order conditions (strong nucleophile), yet the structure of the product shows rearrangement. Also, the rate of this reaction is several thousand times faster than the rate of substitution of hydroxide ion on 2-chlorobutane under similar conditions. Propose a mechanism to explain the enhanced rate and rearrangement observed in this unusual reaction. ("Et" is the abbreviation for ethyl.)



- 6-57** Propose mechanisms to account for the observed products in the following reactions.



- 6-58** Show the products you expect when each compound reacts with NBS with light shining on the reaction.



- 6-59** A student adds NBS to a solution of 1-methylcyclohexene and irradiates the mixture with a sunlamp until all the NBS has reacted. After a careful distillation, the product mixture contains two major products of formula $\text{C}_7\text{H}_{11}\text{Br}$.
- Draw the resonance forms of the three possible allylic free radical intermediates.
 - Rank these three intermediates from most stable to least stable.
 - Draw the products obtained from each free-radical intermediate.